



Submersible Grey Water Pump

AmaDrainer 4 / 5

Type Series Booklet



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Type Series Booklet AmaDrainer 4 / 5

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Drainage

Drainage Pumps / Grey Water Pumps

AmaDrainer 4 / 5



Main applications

- Drainage
- Disposal
- Drainage systems
- Lowering groundwater levels
- Maintaining groundwater levels
- Draining of pits, shafts, etc.
- Surface drainage

Fluids handled

Free passage 10/11 mm:

- Solid particles with a particle size of up to 10 or 11 mm

Free passage 35 mm:

- Grey water containing long fibres and stringy material
- Solid particles with a particle size of up to 35 mm

Material variant A (standard variant)

- Chemically neutral grey water
- Slightly contaminated grey water ($\leq 40^\circ\text{C}$)
- Wash water (up to $\leq 90^\circ\text{C}$ max. for short periods $t \leq 3$ minutes)

Material variant C (for aggressive fluids)

In addition to standard variant:

- Swimming pool water¹⁾
- Brackish water
- Seawater
- Water containing salt
- Aggressive fluids
- Condensate from heat recovery applications

Variant R (for fluids handled containing oil / oil emulsions)

In addition to standard variant:

- Oil emulsions and cutting oils
- Grey water containing oil

Further information on fluids handled

Overview of fluids handled ($\Rightarrow$ Page 8)

Operating data

Table 1: Operating properties

Characteristic		Value
Flow rate	Q [m ³ /h]	≤ 50
	Q [l/s]	≤ 14
Head	H [m]	≤ 24
Fluid temperature	T [°C]	$\leq +40$ (continuous duty)
		$\leq +90$ (short-time duty, ≤ 3 minutes)
Immersion depth	ET [m]	≤ 7

Design details

Design

- Fully floodable submersible motor pump
- Close-coupled design
- Single-stage
- To EN 12050-2
- Vertical discharge nozzle
- With or without level control

Installation

- Vertical installation
- Wet-installed transportable model
- Wet-installed stationary model

Drive

- Motor winding to IEC 60038
- Motor design to EN 60043 T1 / IEC 34-1
- Thermal class F
- DOL starting
- Enclosure IP68 (permanently submerged) to EN 60529 / IEC 529

AmaDrainer NE/SE 10/35

- AC motor
- Integrated temperature switches
- 10-metre power cable
- Shockproof plug

AmaDrainer SD 10/11/35

- Three-phase motor
- Integrated temperature switches
- 10-metre power cable
- CEE plug (3L+PE+N) with motor contactor and phase inverter

1 Swimming pool water (0.4 to 1.4 mg/l free chlorine, combined chlorine ≤ 0.6 mg/l, pH 6.9 to 7.7, water hardness 10 to 30 °dH, salt content ≤ 7 g/l)

AmaDrainer ND 10/11/35

- Three-phase motor
- Integrated temperature switches
- 10-metre power cable with free cable end and protective cap

Shaft seal

- Pump end, 1 bi-directional mechanical seal
- Drive end, 1 shaft seal ring
- Liquid reservoir between the seals for cooling and lubrication

Impeller type

- Open multi-vane impeller
- Vortex impeller

Bearings

- Maintenance-free
- Grease-packed rolling element bearings sealed for life

Designation

Example: AmaDrainer A 4 22 SD / 10 K

Table 2: Designation key

Code	Description	
AmaDrainer	Type series	
A	Material variant	
	A	Standard design
	C	Variant for aggressive fluids
	R	Variant for fluids handled containing oil / oil emulsions
4	Nominal discharge nozzle diameter	
	4	~ 40 mm (G 1 1/2)
	5	~ 50 mm (G 2)
22	Motor rating [kW x 10]	
	05	0,55 kW
	07	0,75 kW
	11	1,1 kW
	15	1,5 kW
	22	2,2 kW
S	Float switch	
	S	With float switch
	N	Without float switch
D	Motor	
	D	Three-phase motor
	E	Single-phase AC motor
10	Free passage [mm]	
	10	10 mm
	11	11 mm
	35	35 mm
K	Cooling jacket	
	K	With cooling jacket
	_2)	Without cooling jacket

Materials

Table 3: Overview of available materials for material variant A

Part No. (⇒ Page 37)	Description	Nominal discharge nozzle diameter [mm]			
		~ 40	~ 40	~ 50	~ 50 ³⁾
		Free passage [mm]			
		10	35	10 ⁴⁾	11
81-45	Float	Polypropylene (PP)			
81-78	Stator case	Chrome nickel steel (1.4301)			
101	Pump casing	Chrome nickel steel (1.4301)			Grey cast iron EN-GJL-250
102	Volute	Acrylonitrile butadiene styrene copolymer (ABS)			-
162	Suction cover	Chrome nickel steel (1.4301)			Grey cast iron EN-GJL-250
182	Pump foot	Polypropylene (PP)	Chrome nickel steel (1.4301)	Polypropylene (PP)	Polyethylene (PE)
210	Shaft	Chrome steel (1.4021)			
230	Impeller	Polyamide (PA)			
433	Mechanical seal	Silicon carbide (SiC/SiC)			
824	Motor power cable	Polychloroprene rubber (CR)			
-	O-rings	Acrylonitrile butadiene rubber (NBR)			
-	Cooling jacket	-	-	Polyoxymethylene (POM)	-
-	Oil supply	Liquid paraffin (environmentally friendly)			

Table 4: Overview of available materials for material variant C

Part No. (⇒ Page 37)	Description	Nominal discharge nozzle diameter [mm]		
		~ 40	~ 50	~ 50 ³⁾
		Free passage [mm]		
		35	10 ⁴⁾	11
81-45	Float	Polypropylene (PP)		
81-78	Stator case	Chrome nickel molybdenum steel (1.4401)		
101	Pump casing	Chrome nickel molybdenum steel (1.4401)		Chrome nickel molybdenum cast steel (1.4408)
102	Volute	Acrylonitrile butadiene styrene copolymer (ABS)		-
162	Suction cover	Chrome nickel molybdenum steel (1.4401)		Chrome nickel molybdenum cast steel (1.4408)
182	Pump foot	Polypropylene (PP)		Polyethylene (PE)
210	Shaft	Chrome nickel molybdenum steel (1.4571)		
230	Impeller	Polyamide (PA)		
433	Mechanical seal	Silicon carbide (SiC/SiC)		
824	Motor power cable	Polychloroprene rubber (CR)		
-	O-rings	Acrylonitrile butadiene rubber (NBR)		
-	Cooling jacket	-	Polyoxymethylene (POM)	-
-	Oil supply	Liquid paraffin (environmentally friendly)		

Table 5: Overview of available materials for material variant R

Part No. (⇒ Page 37)	Description	Nominal discharge nozzle diameter [mm]	
		~ 50	~ 50 ³⁾
		Free passage [mm]	
		10 ⁴⁾	11
81-45	Float	Polypropylene (PP)	
81-78	Stator case	Chrome nickel steel (1.4301)	
101	Pump casing	Chrome nickel steel (1.4301)	Grey cast iron EN-GJL-250
102	Volute	Acrylonitrile butadiene styrene copolymer (ABS)	-
162	Suction cover	Chrome nickel steel (1.4301)	Grey cast iron EN-GJL-250
182	Pump foot	Polypropylene (PP)	
210	Shaft	Chrome steel (1.4021)	

3 Motor rating 2.2 kW

4 With cooling jacket

Part No. (⇒ Page 37)	Description	Nominal discharge nozzle diameter [mm]	
		~ 50	~ 50 ³⁾
		Free passage [mm]	
		10 ⁴⁾	11
230	Impeller	Polyamide (PA)	
433	Mechanical seal	Silicon carbide (SiC/SiC)	
824	Motor power cable	Polyurethane (PUR)	
-	O-rings	Fluorocarbon rubber (FPM)	
-	Cooling jacket	Polyoxymethylene (POM)	-
-	Oil supply	Liquid paraffin (environmentally friendly)	

Product benefits

- Ready-to-connect, easy installation and commissioning
- Reliable shaft sealing by SiC/SiC mechanical seal with good dry-running characteristics, with oil reservoir
- Maintenance-free with grease-packed bearings sealed for life

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Certifications

Table 6: Overview

Label	Effective in:
 Bauart geprüft und überwacht www.tuv.com ID 1111220216	Europe

Overview of product features / selection tables

Overview of fluids handled

The table of fluids handled is a selection aid for different applications. It serves as guidance and is based on long-standing experience. The data are reference values and are not to be considered generally binding recommendations. They shall not be the basis for warranty claims. Please contact your nearest sales branch for in-depth advice.

Table 7: Table of fluids handled

Fluids handled	Temperature	Percentage	Variant				
			A		C		R
			Free passage				
	[°C]	[%]	10/11	35	10/11	35	10/11
Ammonium chloride NH ₄ Cl	-	10	-	-	X	-	-
Ammonium hydroxide NH ₄ OH	≤ 30	10	X	X	-	-	-
Aluminium sulphate Al ₂ (SO ₄) ₃	≤ 40	10	-	-	X	-	-
Antifrogen-water mixture	-	-	X	X	-	-	-
Ethylene glycol	-	-	X	X	-	-	-
Alkaline cleaning agents	-	-	-	-	5)	5)	-
Barium nitrate	-	-	X	X	-	-	-
Drilling emulsion	-	-	-	-	-	-	X
Brackish water	-	-	-	-	X	X	-
Calcium hydroxide Ca (OH) ₂	≤ 30	5	X	X	-	-	-
Calcium chloride CaCl ₂	≤ 25	5	-	-	X	X	-
Calcium nitrate Ca (NO ₃) ₂	-	10	X	X	-	-	-
Deionised water	-	-	X	X	-	-	-
Landfill leachate	-	-	-	-	5)	5)	-
Disinfecting lye	-	-	-	-	5)	5)	-
Iron(II) nitrate Fe (NO ₃) ₂	-	5	-	-	X	-	-
Iron(II) sulphate Fe (SO ₄)	-	5	-	-	X	-	-
Photographic developers	-	-	-	-	5)	-	-
Vinegar	-	-	-	-	X	-	-
Degreaser	-	-	-	-	5)	5)	-
Liquid fertiliser	-	-	-	-	X	X	-
Antifreeze	-	-	X	X	-	-	-
Fibrous material	-	-	-	X	-	X	-
Fruit juice	-	-	-	-	X	X	-
Galvanic baths	-	-	-	-	5)	5)	-
Fermentation juice	-	-	-	-	-	X	-
Ethylene glycol	-	-	X	X	-	-	-
Potassium carbonate	-	-	X	X	-	-	-
Potassium chloride KCl	≤ 20	3	-	-	X	-	-
Potassium hydroxide KOH	≤ 30	10	X	X	-	-	-
Potassium nitrate KNO ₃	-	10	X	X	-	-	-
Calcium hydroxide (lime water) Ca(OH) ₂	≤ 30	5	X	X	-	X	-
Condensate (from condensing boiler applications)	-	-	-	-	5)	-	-
Waste water from laboratories	-	-	-	-	5)	5)	-
Lemonades	-	-	-	-	X	-	-
Magnesium chloride MgCl ₂	≤ 20	3	-	-	X	-	-
Magnesium sulphate MgSO ₄	-	10	X	X	-	-	-
Milk	-	10	X	X	-	-	-
Lactic acids	-	5	-	-	X	-	-
Whey	-	-	-	-	X	-	-
Sodium chloride NaCl	≤ 20	3	-	-	X	-	-
Sodium hydroxide NaOH	≤ 30	10	X	X	-	-	-
Sodium carbonate Na ₂ CO ₃	-	10	X	X	-	-	-
Sodium nitrate	-	-	X	X	-	-	-

5 Contact KSB and provide the relevant analysis as well as the data on temperature and mode of operation.

Fluids handled	Temperature	Percentage	Variant				
			A		C		R
			Free passage				
			10/11	35	10/11	35	10/11
	[°C]	[%]					
Sodium perborate	-	-	X	X	-	-	-
Sodium sulphate Na ₂ SO ₄	-	10	X	X	-	-	-
Oil-water emulsion	-	-	-	-	-	-	X
Osmosis	-	-	-	-	-	-	-
▪ Raw water (containing salt/chloride)	-	-	-	-	5)	5)	-
▪ Permeate (low salt content)	-	-	-	-	X	X	-
Paraffin oil	-	-	-	-	-	-	X
Rapeseed oil	-	-	-	-	-	-	X
Cleaning lye / washing lye	≤ 40	≤ 12 ⁶⁾	-	-	X	X	-
Acids, diluted	≤ 20	≥ 5 ⁶⁾	-	-	X	X	-
Silicone oil	-	-	-	-	-	-	X
Silage leachate	-	-	-	-	X	X	-
Soy-bean oil	-	-	X	X	-	-	X
Edible vinegar	-	-	-	-	X	-	-
Edible oil	-	-	-	-	-	-	X
Cutting oil	-	-	-	-	-	-	X
Trisodium phosphate	-	-	X	X	-	-	-
Vaseline	-	-	X	X	-	-	-
Washing machine lye	-	-	X	X	X	X	-
Lye for metal cleaning	-	-	-	-	5)	5)	-
Water	-	-	-	-	-	-	-
▪ Drainage water	-	-	X	X	-	-	-
▪ Fire-fighting water	-	-	X	X	-	-	-
▪ Heating water	-	-	X	X	-	-	-
▪ Boiler water	-	-	X	X	-	-	-
▪ Cooling water	-	-	X	X	-	-	-
▪ Seawater	≤15	-	-	-	X	X	-
▪ Raw water	-	-	-	-	5)	5)	-
▪ Salt water	-	-	-	-	5)	5)	-
▪ Swimming pool water (DIN 19643)	-	-	-	-	X	X	-
▪ Partly desalinated water	-	-	X	X	-	-	-
▪ Fully desalinated water	-	-	-	-	X	X	-
Waste water	-	-	-	-	-	-	-
▪ Electroplating shops	-	-	-	-	5)	5)	-
▪ Bottle washing, crate washing, keg washing	-	-	-	-	X	X	-
▪ Beverage industry, breweries	-	-	-	-	X	X	-
▪ Dairies, winegrowers' cooperatives	-	-	-	-	X	X	-
▪ Emergency pumping (floods)	-	-	-	X	-	X	-
▪ Containing salt (from fish-processing plants)	-	-	-	-	-	X	-
▪ Seawater and river water	-	-	-	X	-	X	-
▪ Car repair shops, car washes	-	-	-	-	-	-	X
▪ Petrol stations	-	-	-	-	-	-	X
▪ Pit drainage (raffinates)	-	-	-	-	-	-	X

Fluids handled	Temperature	Percentage	Variant				
			A		C		R
			Free passage				
	[°C]	[%]	10/11	35	10/11	35	10/11
▪ Pit drainage (chemically aggressive substances)	-	-	-	-	X	X	-
▪ Wash water containing long fibres and stringy material	-	-	-	X	-	X	-
Citric acid	≤ 10	-	-	-	X	-	-

Technical data

AmaDrainer 4 / 5, variant A for grey water, without cooling jacket (standard design)

ND = three-phase asynchronous motor without float switch

NE = single phase AC motor without float switch

SD = three-phase asynchronous motor with float switch

SE = single phase AC motor with float switch

Table 8: 50 Hz

AmaDrainer	Discharge-side connection	Free pas- sage	P ₁	P _N	I _N		Power cable				Level control		Mat. No.	[kg]
					1~230 V	3~400 V	H07RN-F6G1		H07RN-F3G1		H07RN-F3G1			
							[A]	[A]	[m]	[mm²]	[m]	[mm²]		
A 405 NE/10	G 1 1/2	10	0,90	0,55	3,50	-	-	-	10	3 × 1,0	-	-	29128651	12.2
A 405 SE/10	G 1 1/2	10	0,90	0,55	3,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128650	12.7
A 405 ND/10	G 1 1/2	10	0,76	0,55	-	1,70	10	6 × 1,0	-	-	-	-	29128652	13.8
A 405 SD/10	G 1 1/2	10	0,76	0,55	-	1,70	10	6 × 1,0	-	-	10	3 × 1,0	29128742	15.1
A 405 NE/35	G 1 1/2	35	0,90	0,55	3,50	-	-	-	10	3 × 1,0	-	-	29128677	13.2
A 405 SE/35	G 1 1/2	35	0,90	0,55	3,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128676	13.7
A 405 ND/35	G 1 1/2	35	0,76	0,55	-	1,70	10	6 × 1,0	-	-	-	-	29128678	14.8
A 405 SD/35	G 1 1/2	35	0,76	0,55	-	1,70	10	6 × 1,0	-	-	10	3 × 1,0	29128752	16.1
A 407 NE/10	G 1 1/2	10	1,26	0,75	4,50	-	-	-	10	3 × 1,0	-	-	29128654	12.2
A 407 SE/10	G 1 1/2	10	1,26	0,75	4,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128653	12.7
A 407 ND/10	G 1 1/2	10	1,01	0,75	-	1,90	10	6 × 1,0	-	-	-	-	29128655	13.8
A 407 SD/10	G 1 1/2	10	1,01	0,75	-	1,90	10	6 × 1,0	-	-	10	3 × 1,0	29128743	15.1
A 411 NE/10	G 1 1/2	10	1,45	1,10	6,85	-	-	-	10	3 × 1,0	-	-	29128657	14.5
A 411 SE/10	G 1 1/2	10	1,45	1,10	6,85	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128656	15
A 411 ND/10	G 1 1/2	10	1,54	1,10	-	2,50	10	6 × 1,0	-	-	-	-	29128658	13.8
A 411 SD/10	G 1 1/2	10	1,54	1,10	-	2,50	10	6 × 1,0	-	-	10	3 × 1,0	29128744	15.1
A 411 NE/35	G 1 1/2	35	1,45	1,10	6,85	-	-	-	10	3 × 1,0	-	-	29128680	15.5
A 411 SE/35	G 1 1/2	35	1,45	1,10	6,85	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128679	16
A 411 ND/35	G 1 1/2	35	1,54	1,10	-	2,50	10	6 × 1,0	-	-	-	-	29128681	14.8
A 411 SD/35	G 1 1/2	35	1,54	1,10	-	2,50	10	6 × 1,0	-	-	10	3 × 1,0	29128753	16.1
A 415 NE/10	G 1 1/2	10	2,07	1,50	9,60	-	-	-	10	3 × 1,0	-	-	29128660	14.5
A 415 SE/10	G 1 1/2	10	2,07	1,50	9,60	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128659	15
A 415 ND/10	G 1 1/2	10	1,88	1,50	-	3,60	10	6 × 1,0	-	-	-	-	29128661	15.6
A 415 SD/10	G 1 1/2	10	1,88	1,50	-	3,60	10	6 × 1,0	-	-	10	3 × 1,0	29128745	16.9
A 422 ND/10	G 1 1/2	10	2,90	2,20	-	4,80	10	6 × 1,0	-	-	-	-	29128662	15.6
A 422 SD/10	G 1 1/2	10	2,90	2,20	-	4,80	10	6 × 1,0	-	-	10	3 × 1,0	29128746	16.9
A 422 ND/35	G 1 1/2	35	2,90	2,20	-	4,80	10	6 × 1,0	-	-	-	-	29128682	16.6
A 422 SD/35	G 1 1/2	35	2,90	2,20	-	4,80	10	6 × 1,0	-	-	10	3 × 1,0	29128754	17.9
A 522 ND/11	G 2	11	2,90	2,20	-	4,80	10	6 × 1,0	-	-	-	-	29128865	25
A 522 SD/11	G 2	11	2,90	2,20	-	4,80	10	6 × 1,0	-	-	10	3 × 1,0	29128866	27

AmaDrainer 4 / 5, variant A for grey water, with cooling jacket (standard design)

ND = three-phase asynchronous motor without float switch

NE = single phase AC motor without float switch

SD = three-phase asynchronous motor with float switch

SE = single phase AC motor with float switch

Table 9: 50 Hz

AmaDrainer	Discharge-side connection	Free pas- sage	P ₁	P _N	I _N		Power cable				Level control		Mat. No.	[kg]
					1-230 V	3~400 V	H07RN-F6G1		H07RN-F3G1		H07RN-F3G1			
							[A]	[A]	[m]	[mm ²]	[m]	[mm ²]		
A 505 NE/10K	G 2	10	0,90	0,55	3,50	-	-	-	10	3 × 1,0	-	-	29128664	14.2
A 505 SE/10K	G 2	10	0,90	0,55	3,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128663	14.7
A 505 ND/10K	G 2	10	0,76	0,55	-	1,70	10	6 × 1,0	-	-	-	-	29128665	15.8
A 505 SD/10K	G 2	10	0,76	0,55	-	1,70	10	6 × 1,0	-	-	10	3 × 1,0	29128747	17.1
A 507 NE/10K	G 2	10	1,26	0,75	4,50	-	-	-	10	3 × 1,0	-	-	29128667	14.2
A 507 SE/10K	G 2	10	1,26	0,75	4,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128666	14.7
A 507 ND/10K	G 2	10	1,01	0,75	-	1,90	10	6 × 1,0	-	-	-	-	29128668	15.8
A 507 SD/10K	G 2	10	1,01	0,75	-	1,90	10	6 × 1,0	-	-	10	3 × 1,0	29128748	17.1
A 511 NE/10K	G 2	10	1,45	1,10	6,85	-	-	-	10	3 × 1,0	-	-	29128670	16.5
A 511 SE/10K	G 2	10	1,45	1,10	6,85	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128669	17
A 511 ND/10K	G 2	10	1,54	1,10	-	2,50	10	6 × 1,0	-	-	-	-	29128671	15.8
A 511 SD/10K	G 2	10	1,54	1,10	-	2,50	10	6 × 1,0	-	-	10	3 × 1,0	29128749	17.1
A 515 NE/10K	G 2	10	2,07	1,50	9,60	-	-	-	10	3 × 1,0	-	-	29128673	16.5
A 515 SE/10K	G 2	10	2,07	1,50	9,60	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128672	17
A 515 ND/10K	G 2	10	1,88	1,50	-	3,60	10	6 × 1,0	-	-	-	-	29128674	17.6
A 515 SD/10K	G 2	10	1,88	1,50	-	3,60	10	6 × 1,0	-	-	10	3 × 1,0	29128750	18.9
A 522 ND/10K	G 2	10	2,90	2,20	-	4,80	10	6 × 1,0	-	-	-	-	29128675	17.6
A 522 SD/10K	G 2	10	2,90	2,20	-	4,80	10	6 × 1,0	-	-	10	3 × 1,0	29128751	18.9

AmaDrainer 4 / 5, variant C for aggressive water, without cooling jacket

ND = three-phase asynchronous motor without float switch

NE = single phase AC motor without float switch

SD = three-phase asynchronous motor with float switch

SE = single phase AC motor with float switch

Table 10: 50 Hz

AmaDrainer	Discharge-side connection	Free pas- sage	P ₁	P _N	I _N		Power cable				Level control		Mat. No.	[kg]
					1~230 V	3~400 V	H07RN-F6G1		H07RN-F3G1		H07RN-F3G1			
							[A]	[A]	[m]	[mm ²]	[m]	[mm ²]		
C 405 NE/35	G 1 1/2	35	0,90	0,55	3,50	-	-	-	10	3 × 1,0	-	-	29128683	15.3
C 405 SE/35	G 1 1/2	35	0,90	0,55	3,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128684	15.8
C 405 ND/35	G 1 1/2	35	0,76	0,55	-	1,70	10	6 × 1,0	-	-	-	-	29128685	16.5
C 405 SD/35	G 1 1/2	35	0,76	0,55	-	1,70	10	6 × 1,0	-	-	10	3 × 1,0	29128686	18.2
C 411 NE/35	G 1 1/2	35	1,45	1,10	6,85	-	-	-	10	3 × 1,0	-	-	29128687	17.6
C 411 SE/35	G 1 1/2	35	1,45	1,10	6,85	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128688	18.1
C 411 ND/35	G 1 1/2	35	1,54	1,10	-	2,50	10	6 × 1,0	-	-	-	-	29128689	16.5
C 411 SD/35	G 1 1/2	35	1,54	1,10	-	2,50	10	6 × 1,0	-	-	10	3 × 1,0	29128690	18.3
C 422 ND/35	G 1 1/2	35	2,90	2,20	-	4,80	10	6 × 1,0	-	-	-	-	29128691	19
C 422 SD/35	G 1 1/2	35	2,90	2,20	-	4,80	10	6 × 1,0	-	-	10	3 × 1,0	29128692	20.8
C 522 ND/11	G 2	11	2,90	2,20	-	4,80	10	6 × 1,0	-	-	-	-	29128693	23.5
C 522 SD/11	G 2	11	2,90	2,20	-	4,80	10	6 × 1,0	-	-	10	3 × 1,0	29128694	25.5

AmaDrainer 4 / 5, variant C for aggressive water, with cooling jacket

ND = three-phase asynchronous motor without float switch

NE = single phase AC motor without float switch

SD = three-phase asynchronous motor with float switch

SE = single phase AC motor with float switch

Table 11: 50 Hz

AmaDrainer	Discharge-side connection	Free pas- sage	P ₁	P _N	I _N		Power cable				Level control		Mat. No.	[kg]
					1~230 V	3~400 V	H07RN-F6G1		H07RN-F3G1		H07RN-F3G1			
							[m]	[mm ²]	[m]	[mm ²]	[m]	[mm ²]		
C 505 NE/10K	G 2	10	0,90	0,55	3,50	-	-	-	10	3 × 1,0	-	-	29128697	14.2
C 505 SE/10K	G 2	10	0,90	0,55	3,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128696	14.7
C 505 ND/10K	G 2	10	0,76	0,55	-	1,70	10	6 × 1,0	-	-	-	-	29128698	15.3
C 505 SD/10K	G 2	10	0,76	0,55	-	1,70	10	6 × 1,0	-	-	10	3 × 1,0	29128755	17.1
C 507 NE/10K	G 2	10	1,26	0,75	4,50	-	-	-	10	3 × 1,0	-	-	29128700	14.2
C 507 SE/10K	G 2	10	1,26	0,75	4,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128699	14.7
C 507 ND/10K	G 2	10	1,01	0,75	-	1,90	10	6 × 1,0	-	-	-	-	29128701	15.3
C 507 SD/10K	G 2	10	1,01	0,75	-	1,90	10	6 × 1,0	-	-	10	3 × 1,0	29128756	17.1
C 511 NE/10K	G 2	10	1,45	1,10	6,85	-	-	-	10	3 × 1,0	-	-	29128703	16.5
C 511 SE/10K	G 2	10	1,45	1,10	6,85	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128702	17
C 511 ND/10K	G 2	10	1,54	1,10	-	2,50	10	6 × 1,0	-	-	-	-	29128704	15.3
C 511 SD/10K	G 2	10	1,54	1,10	-	2,50	10	6 × 1,0	-	-	10	3 × 1,0	29128757	17.1
C 515 NE/10K	G 2	10	2,07	1,50	9,60	-	-	-	10	3 × 1,0	-	-	29128706	16.5
C 515 SE/10K	G 2	10	2,07	1,50	9,60	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128705	17
C 515 ND/10K	G 2	10	1,88	1,50	-	3,60	10	6 × 1,0	-	-	-	-	29128707	17.6
C 515 SD/10K	G 2	10	1,88	1,50	-	3,60	10	6 × 1,0	-	-	10	3 × 1,0	29128758	19.5
C 522 ND/10K	G 2	10	2,90	2,20	-	4,80	10	6 × 1,0	-	-	-	-	29128708	17.7
C 522 SD/10K	G 2	10	2,90	2,20	-	4,80	10	6 × 1,0	-	-	10	3 × 1,0	29128759	19.6

AmaDrainer 4 / 5, variant R for water containing oil / oil emulsions, without cooling jacket

ND = three-phase asynchronous motor without float switch

SD = three-phase asynchronous motor with float switch

Table 12: 50 Hz

AmaDrainer	Discharge-side connection	Free pas- sage	P ₁	P _N	I _N		Power cable				Level control		Mat. No.	[kg]
					1-230 V	3-400 V	PUR 6 x 1		PUR 3 x 1		PUR 3 x 1			
					[mm]	[kW]	[kW]	[A]	[A]	[m]	[mm²]	[m]		
R 522 ND/11	G 2	11	2,90	2,20	-	4,80	10	6 x 1,0	-	3 x 1,0	-	3 x 1,0	29128867	23.5
R 522 SD/11	G 2	11	2,90	2,20	-	4,80	10	6 x 1,0	-	3 x 1,0	10	3 x 1,0	29128868	25.5

AmaDrainer 4 / 5, variant R for water containing oil / oil emulsions, with cooling jacket

ND = three-phase asynchronous motor without float switch

NE = single phase AC motor without float switch

SD = three-phase asynchronous motor with float switch

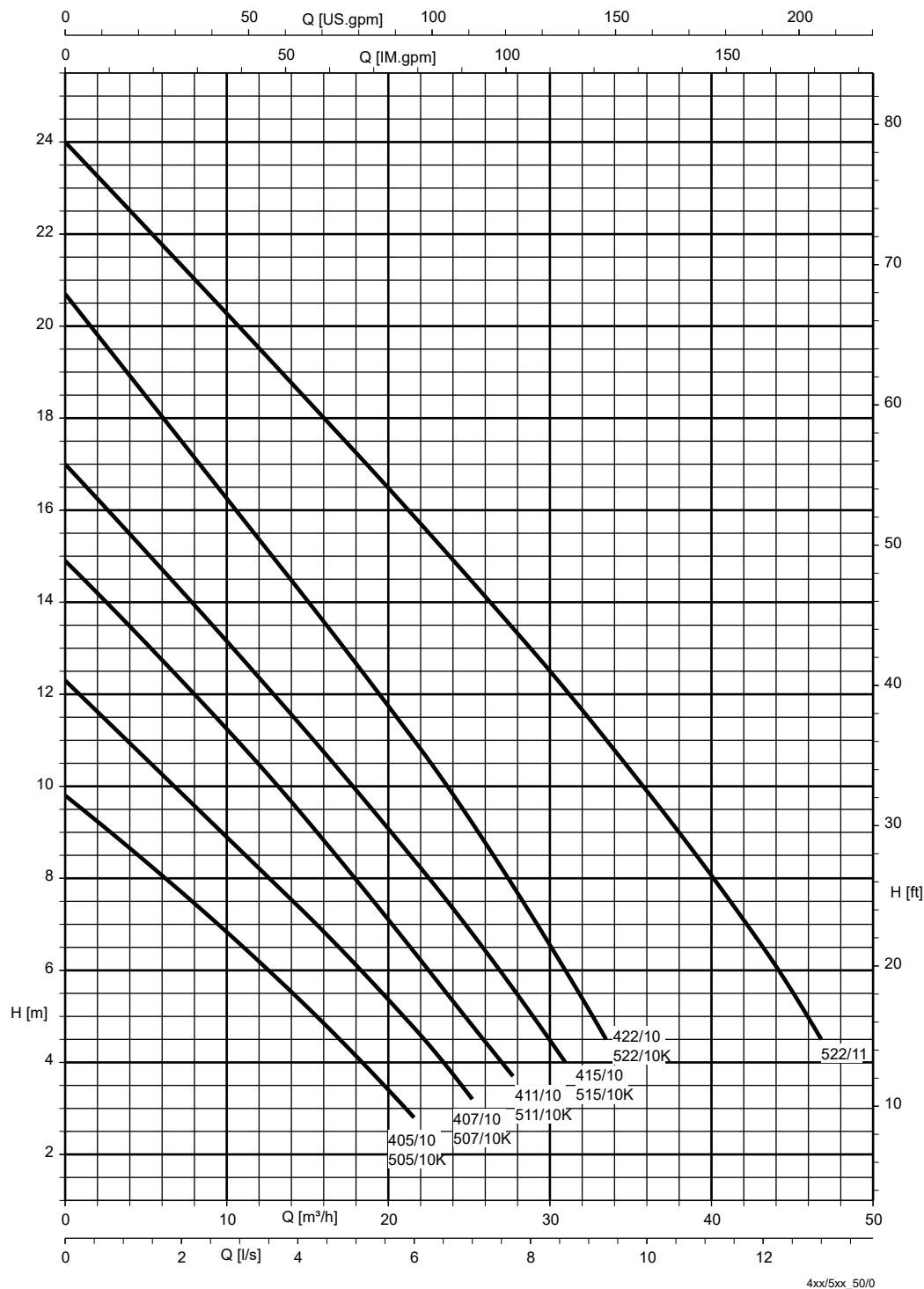
SE = single phase AC motor with float switch

Table 13: 50 Hz

AmaDrainer	Discharge-side connection	Free pas- sage	P ₁	P _N	I _N		Power cable				Level control		Mat. No.	[kg]
					1~230 V	3~400 V	PUR 6 x 1		PUR 3 x 1		PUR 3 x 1			
							[A]	[A]	[m]	[mm²]	[m]	[mm²]		
R 505 NE/10K	G 2	10	0,90	0,55	3,50	-	-	-	10	3 × 1,0	-	-	29128723	14.2
R 505 SE/10K	G 2	10	0,90	0,55	3,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128722	14.7
R 505 ND/10K	G 2	10	0,76	0,55	-	1,70	10	6 × 1,0	-	-	-	-	29128724	15.3
R 505 SD/10K	G 2	10	0,76	0,55	-	1,70	10	6 × 1,0	-	-	10	3 × 1,0	29128760	17.1
R 507 NE/10K	G 2	10	1,26	0,75	4,50	-	-	-	10	3 × 1,0	-	-	29128726	14.2
R 507 SE/10K	G 2	10	1,26	0,75	4,50	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128725	14.7
R 507 ND/10K	G 2	10	1,01	0,75	-	1,90	10	6 × 1,0	-	-	-	-	29128727	15.3
R 507 SD/10K	G 2	10	1,01	0,75	-	1,90	10	6 × 1,0	-	-	10	3 × 1,0	29128761	17.1
R 511 NE/10K	G 2	10	1,45	1,10	6,85	-	-	-	10	3 × 1,0	-	-	29128729	16.5
R 511 SE/10K	G 2	10	1,45	1,10	6,85	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128728	17
R 511 ND/10K	G 2	10	1,54	1,10	-	2,50	10	6 × 1,0	-	-	-	-	29128730	15.3
R 511 SD/10K	G 2	10	1,54	1,10	-	2,50	10	6 × 1,0	-	-	10	3 × 1,0	29128762	17.1
R 515 NE/10K	G 2	10	2,07	1,50	9,60	-	-	-	10	3 × 1,0	-	-	29128732	16.5
R 515 SE/10K	G 2	10	2,07	1,50	9,60	-	-	-	10	3 × 1,0	0,5	3 × 1,0	29128731	17
R 515 ND/10K	G 2	10	1,88	1,50	-	3,60	10	6 × 1,0	-	-	-	-	29128733	17.6
R 515 SD/10K	G 2	10	1,88	1,50	-	3,60	10	6 × 1,0	-	-	10	3 × 1,0	29128763	19.5
R 522 ND/10K	G 2	10	2,90	2,20	-	4,80	10	6 × 1,0	-	-	-	-	29128734	17.7
R 522 SD/10K	G 2	10	2,90	2,20	-	4,80	10	6 × 1,0	-	-	10	3 × 1,0	29128764	19.6

Characteristic curves

AmaDrainer 4 / 5, n = 2800 rpm, multi-vane impeller



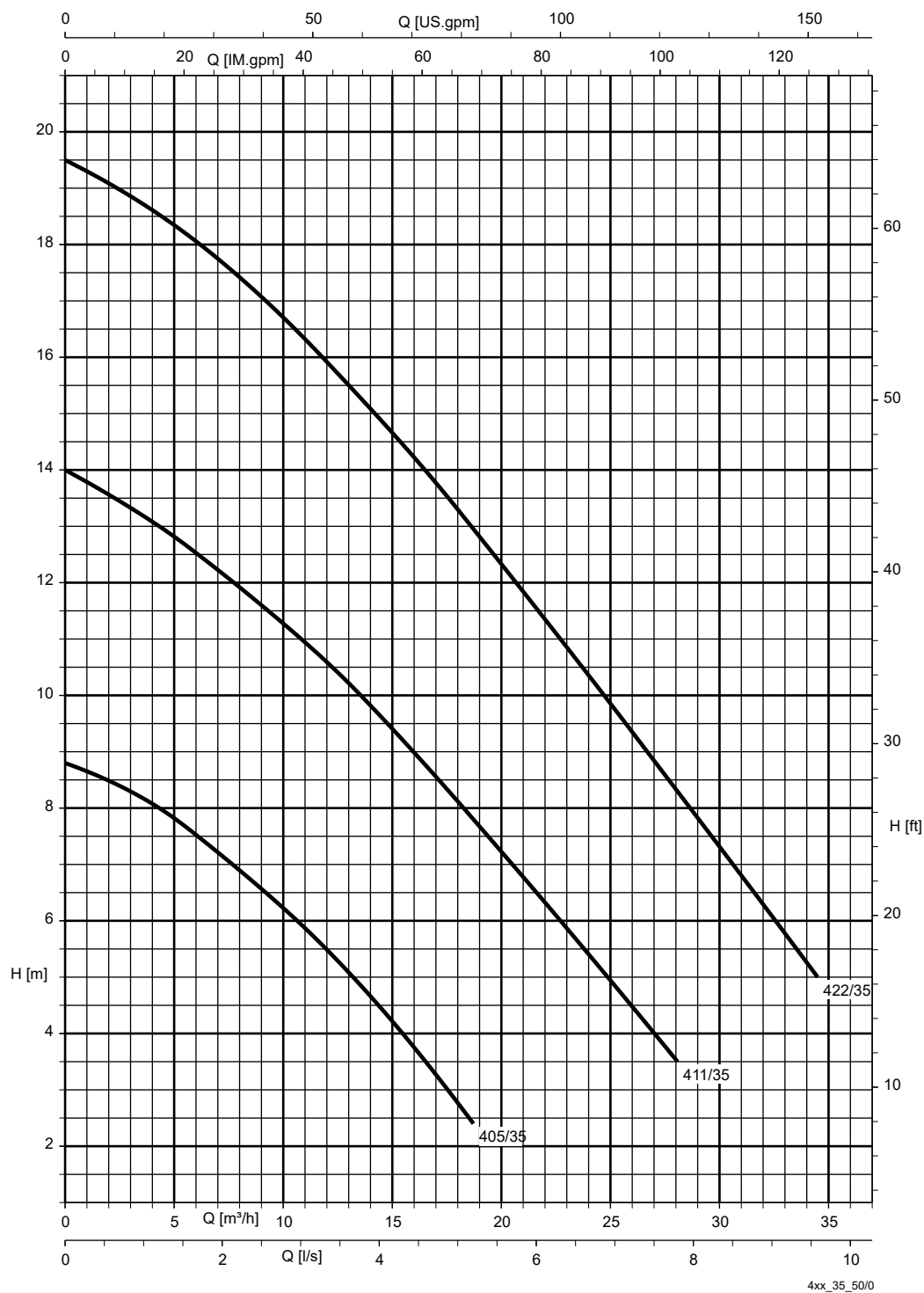
Sizes 405, 407, 411, 415, 422, 505, 507, 511, 515: free passage = 10 mm

Size 522: free passage = 11 mm

Performance tolerance to ISO 2548 Class C (water under standard conditions)

The characteristic curves shown are those of the respective pumps. Hydraulic losses caused by the discharge-side piping, both inside and outside of the AmaDrainer Box lifting unit, must be taken into account during product selection.

AmaDrainer 4, n = 2800 rpm, F impeller



Sizes 405, 411, 422: free passage = 35 mm

Performance tolerance to ISO 2548 Class C (water under standard conditions)

The characteristic curves shown are those of the respective pumps. Hydraulic losses caused by the discharge-side piping, both inside and outside of the AmaDrainer Box lifting unit, must be taken into account during product selection.

Dimensions

AmaDrainer 4, free passage 10 mm, without cooling jacket

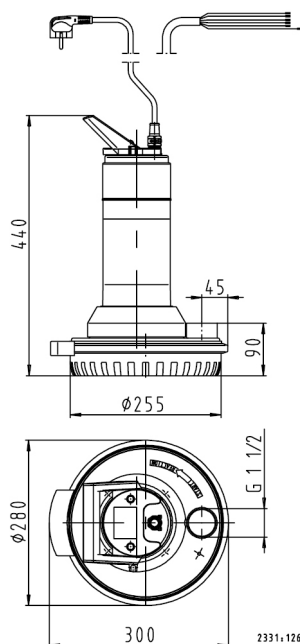


Fig. 1: AmaDrainer 4, free passage 10 mm, without cooling jacket

AmaDrainer 4, free passage 35 mm, without cooling jacket

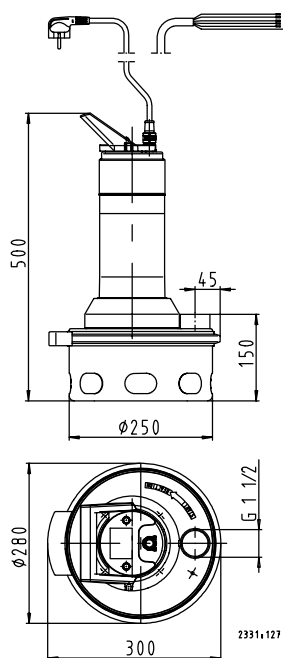


Fig. 2: AmaDrainer 4, free passage 35 mm, without cooling jacket

AmaDrainer 5, free passage 10 mm, with cooling jacket

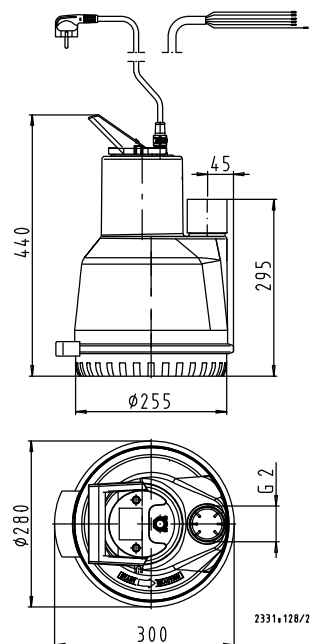


Fig. 3: AmaDrainer 5, free passage 10 mm, with cooling jacket

AmaDrainer 5, free passage 11 mm, without cooling jacket

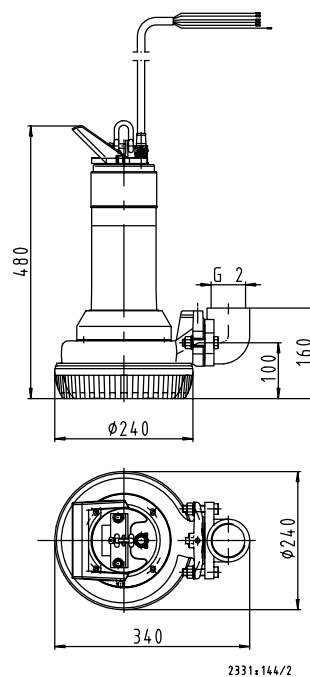


Fig. 4: AmaDrainer 5, free passage 11 mm, without cooling jacket

AmaDrainer 4, single pump, free passage 10 mm, with float switch, single-phase AC motor, without cooling jacket

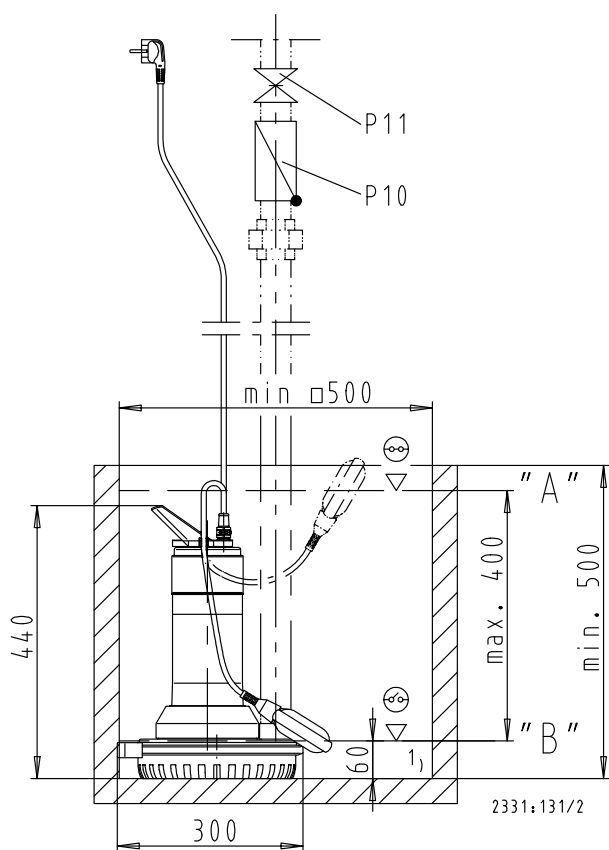


Fig. 5: AmaDrainer 4, single pump, free passage 10 mm, with float switch, single-phase AC motor, without cooling jacket

1)	Residual water level
A	Start-up level
B	Stop level

Pos.	Description
P10	Swing check valve
P11	Gate valve

AmaDrainer 4, single pump, free passage 35 mm, with float switch, three-phase motor, without cooling jacket

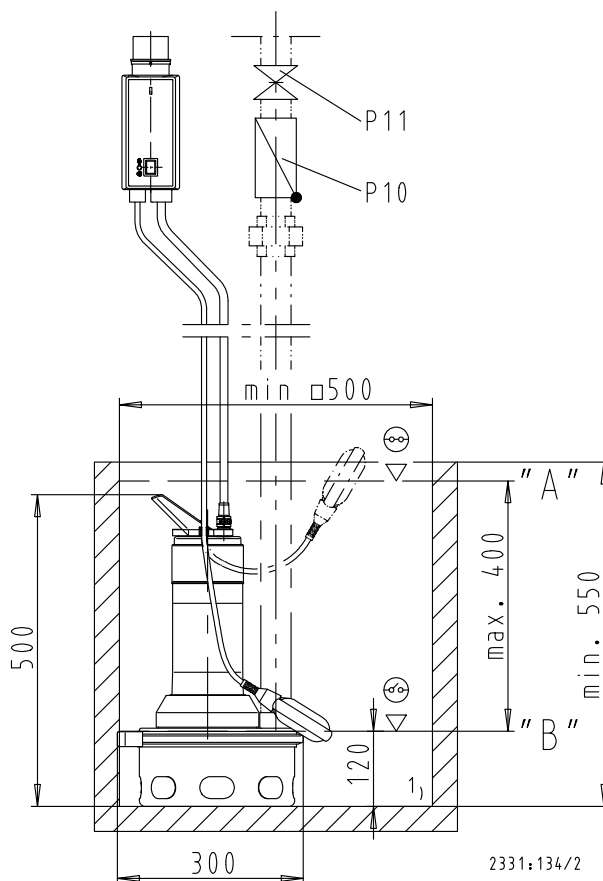


Fig. 6: AmaDrainer 4, single pump, free passage 35 mm, with float switch, three-phase motor, without cooling jacket

1)	Residual water level
A	Start-up level
B	Stop level

Pos.	Description
P10	Swing check valve
P11	Gate valve

AmaDrainer 5, single pump, free passage 10 mm, with float switch, three-phase motor, with cooling jacket

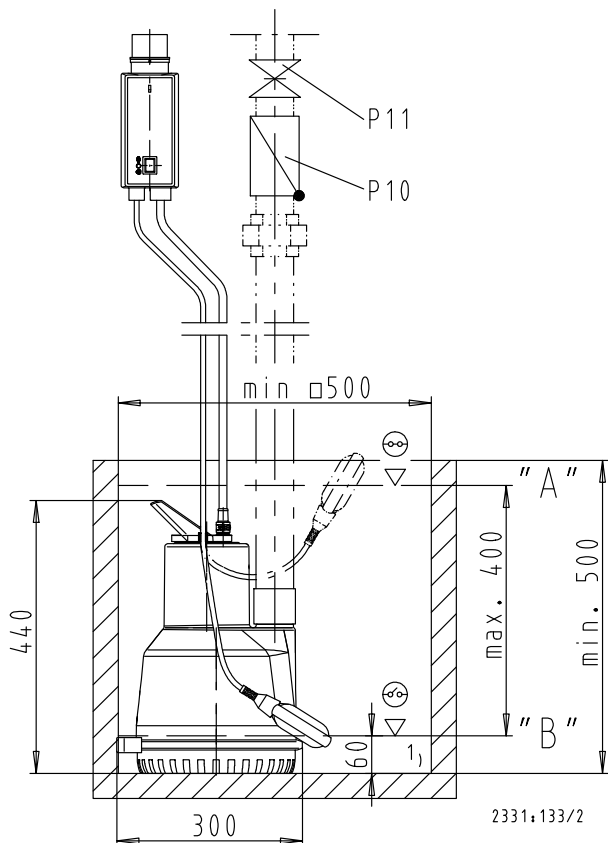


Fig. 7: AmaDrainer 5, single pump, free passage 10 mm, with float switch, three-phase motor, with cooling jacket

1)	Residual water level
A	Start-up level
B	Stop level

Pos.	Description
P10	Swing check valve
P11	Gate valve

AmaDrainer 5, single pump, free passage 11 mm, with float switch, three-phase motor, without cooling jacket

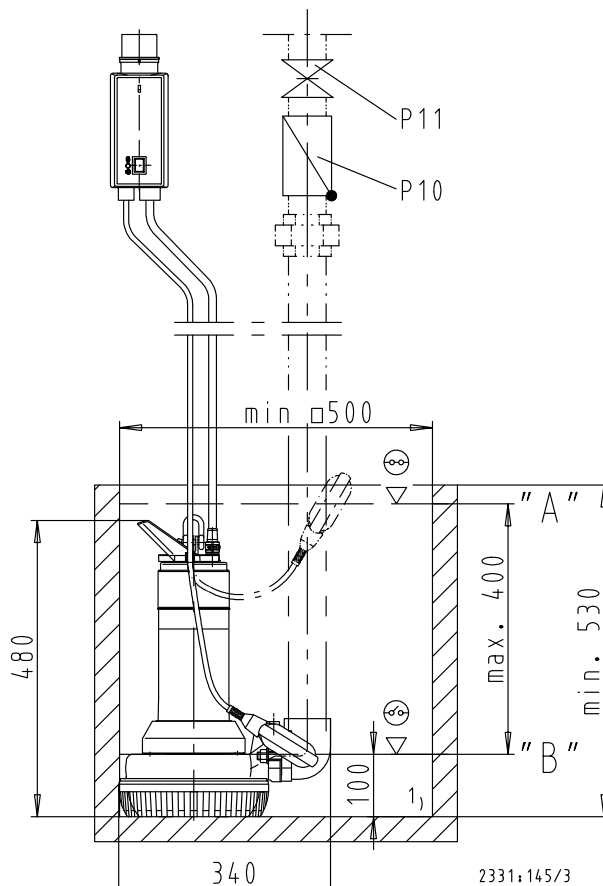


Fig. 8: AmaDrainer 5, single pump, free passage 11 mm, with float switch, three-phase motor, without cooling jacket

1)	Residual water level
A	Start-up level
B	Stop level

Pos.	Description
P10	Swing check valve
P11	Gate valve

AmaDrainer 4, transportable model, free passage 10 mm, without float switch, single-phase AC motor, without cooling jacket

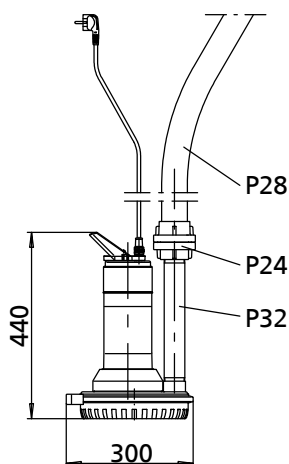


Fig. 9: AmaDrainer 4, transportable model, free passage 10 mm, without float switch, single-phase AC motor, without cooling jacket

Pos.	Description
P24	Storz rigid coupling
P28	Synthetic hose
P32	Pipe extension

AmaDrainer 5, transportable model, free passage 11 mm, without float switch, three-phase motor, without cooling jacket

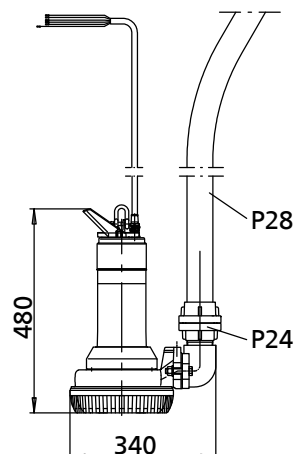


Fig. 11: AmaDrainer 5, transportable model, free passage 11 mm, without float switch, three-phase motor, without cooling jacket

Pos.	Description
P24	Storz rigid coupling
P28	Synthetic hose

AmaDrainer 5, transportable model, free passage 10 mm, without float switch, single-phase AC motor, with cooling jacket

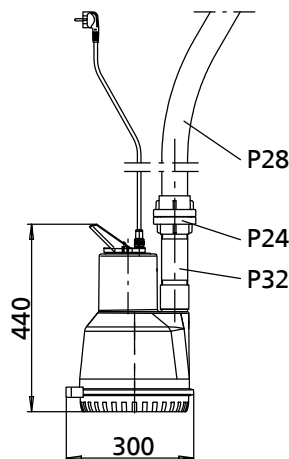


Fig. 10: AmaDrainer 5, transportable model, free passage 10 mm, without float switch, single-phase AC motor, with cooling jacket

Pos.	Description
P24	Storz rigid coupling
P28	Synthetic hose
P32	Pipe extension

AmaDrainer 5, stationary installation with guide hoop, free passage 11 mm, without float switch, three-phase motor, without cooling jacket

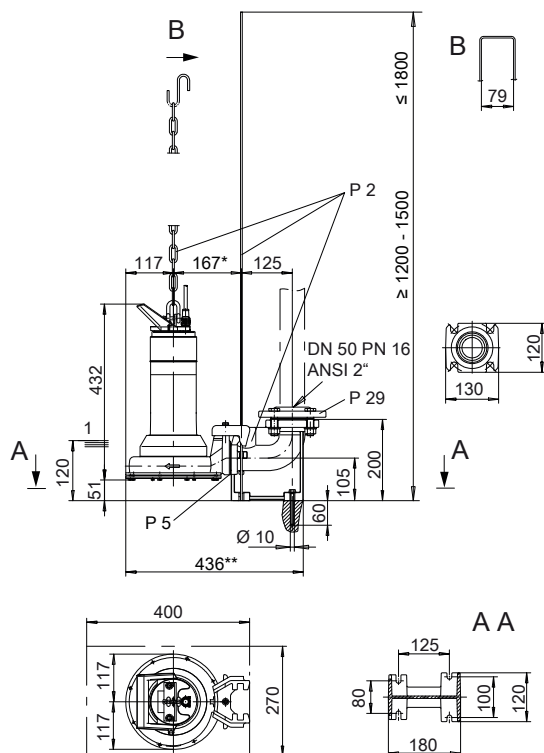


Fig. 12: AmaDrainer 5, stationary installation with guide hoop, free passage 11 mm, without float switch, three-phase motor, without cooling jacket

*	When using flange adapter P5: 217 mm
**	When using flange adapter P5: 486 mm
1	Lowest stop level for automatic operation

Table 14: Connections

Pos.	Description
P2	Guide hoop arrangement
P5 ⁷⁾	Flange adapter for stabilising the pump position during start-up
P29	Threaded flange

AmaDrainer 5, stationary installation with guide hoop, free passage 11 mm, without float switch, three-phase motor, without cooling jacket

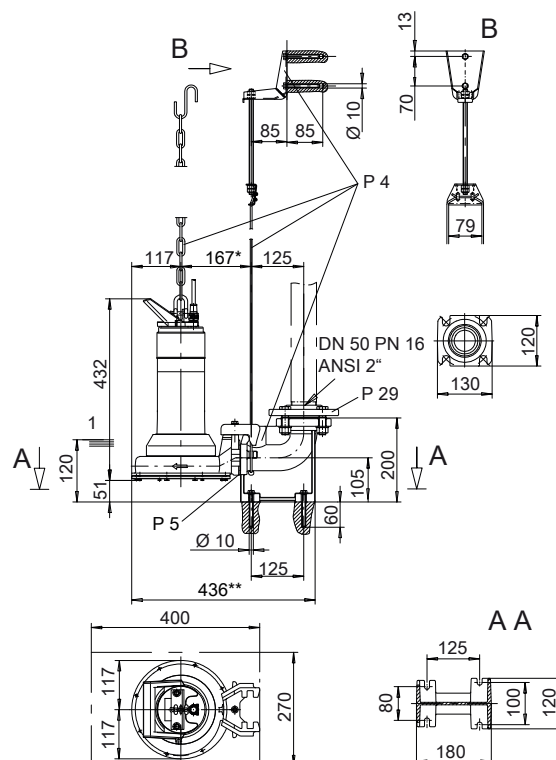


Fig. 13: AmaDrainer 5, stationary installation with guide hoop, free passage 11 mm, without float switch, three-phase motor, without cooling jacket

*	When using flange adapter P5: 217 mm
**	When using flange adapter P5: 486 mm
1	Lowest stop level for automatic operation

Table 15: Connections

Pos.	Description
P4	Guide wire arrangement
P5 ⁸⁾	Flange adapter for stabilising the pump position during start-up
P29	Threaded flange

⁷ Not shown in drawing

⁸ Not shown in drawing

Installation information

Installation example of a dual-pump station

The inlet is positioned above the highest float switch.

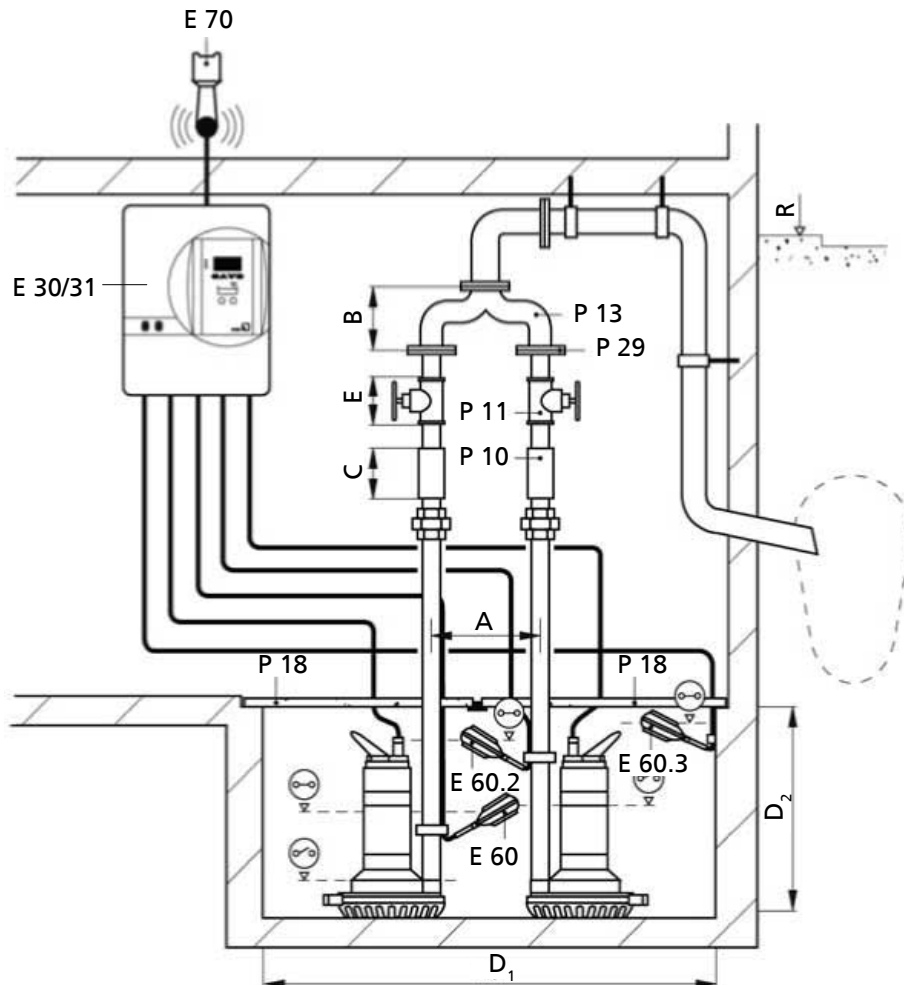


Fig. 14: Position of float switches in a dual-pump station

E70	Horn	P11	Gate valve
E30/E31	Control unit/switchgear	P13	Y-pipe
E60	Float switch, normal water level	P18	Cover plate
E60.2	High water float switch	P29	Threaded flange
E60.3	Alarm contactor	R	Flood level
P10	Swing check valve		

Table 16: Dimensions and weight

Size	A	B	C	D ₁	D ₂	E	[kg]
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	
AmaDrainer 4../10	275	190	130	1060 × 500	500	55	16
AmaDrainer 4../35	275	190	130	1060 × 500	500	60	17
AmaDrainer 5../10 K	300	210	130	1060 × 500	500	55	17
AmaDrainer 522/11	300	210	130	1060 × 500	500	55	24

Accessories
Pump accessories
Table 17: Overview of pump accessories

Item	Description	Connection	Immersion depth [m]	AmaDrainer				Mat. No.	[kg]
				4./10	4./35	5./10K	522/11		
P2 + P5	Installation parts for stationary wet installation (guide hoop arrangement)	-	1,0	-	-	-	X	39020769	11
	Grey cast iron, comprising: DN 50 flanged duckfoot bend, guide hoop, claw with stainless steel screws/bolts	-	1,5	-	-	-	X	39020770	12
		-	1,8	-	-	-	X	39020771	13
	Flange adapter, grey cast iron (must be specially ordered)	DN 50	-	-	-	-	X	19075508	4.2
P4 + P5	Installation parts for stationary wet installation (guide wire arrangement)	-	4,5	-	-	-	X	39021023	10.6
	Grey cast iron, comprising: duckfoot bend, suspension bracket, mounting bracket, 10 m guide wire, claw with stainless steel screws/bolts								
	Flange adapter, grey cast iron (must be specially ordered)	DN 50	-	-	-	-	X	19075508	4.2
	Installation parts for stationary wet installation (guide wire arrangement)	-	4,5	-	-	-	X	19552258	11.83
	Material variant C, stainless steel 1.4571/1.4571, comprising: duckfoot bend, suspension bracket, mounting bracket, 10 m guide wire, screws/bolts and fixings								
P7	Chain 1.0038+Z, hook 1.4571 and shackle 1.4401	2 m, B5 x 35	-	-	-	-	X	19141819	1.5
		5 m, B5/6	-	-	-	-	X	19141820	2.8
	Chain (1.4404) short-linked, tested and duly labelled to Directive 2006/42/EC, shackle (1.4404), hook (1.4301)	2 m, 4 x 16	-	-	-	-	X	01236267	0.9
	Chain (1.4404) short-linked, tested and duly labelled to Directive 2006/42/EC, shackle (1.4404), hook (1.4301)	3 m, 4 x 16	-	-	-	-	X	01236268	1.09
	Chain (1.4404) short-linked, tested and duly labelled to Directive 2006/42/EC, shackle (1.4404), hook (1.4301)	5 m, 4 x 16	-	-	-	-	X	01236269	1.69
	Polypropylene lifting rope, material variant C	5 m	-	-	-	-	X	11185207	2
P8	Flange for pipe coupling PN 10, at the flanged bend, mating dimensions to PN 16	DN 50 / R 2	-	-	-	-	X	19551111	1.2
P10	RK swing check valve	Rp 1 1/4	-	X ^(*)	-	X ^(*)	-	01009771	0.1
	Plastic, EN 12050-4, with internal thread ISO 7/1, with full bore and drain plug	Rp 1 1/2	-	X	X	X ^(*)	-	01009772	0.52
	For additional parts see P32 and P33	Rp 2	-	X ^(*)	X ^(*)	X	X	01009773	0.5
P10	Lift check valve, stainless steel (1.4401)	Rp 1 1/4	-	X ^(*)	-	X ^(*)	-	01084936	2.1
	For additional parts see P32 and P33	Rp 1 1/2	-	X	X	X ^(*)	-	01084935	2.2
		Rp 2	-	X ^(*)	X ^(*)	X	X	01084937	0.44
P11	Socket gate valve	Rp 1 1/4	-	X ^(*)	-	X ^(*)	-	01014219	0.63
	Material: CuZn, PN 16, with internal thread, with full bore	Rp 1 1/2	-	X	X	X ^(*)	-	00411502	0.78
	For additional parts see P32 and P33	Rp 2	-	X ^(*)	X ^(*)	X	X	00411503	1.29
P11	Ball valve	Rp 1 1/4	-	X ^(*)	-	X ^(*)	-	01067465	1.21
	Stainless steel (1.4408), PN 10	Rp 1 1/2	-	X	X	X ^(*)	-	01087276	1.82
		Rp 2	-	X ^(*)	X ^(*)	X	X	01064013	2.1
P13	Y-pipe	Rp 1 1/4	-	X ^(*)	-	-	-	18040311	4.1
P13	For dual-pump sets, with external thread, galvanised steel								
	For additional parts see P32 and P33								
P13	For dual-pump sets, grey cast iron, with hexagon head bolts, nuts and gaskets; flanges drilled to DIN 2501	DN 40	-	X	X	X ^(*)	-	40000688	10.6
	Additional part – see P29	DN 50	-	X ^(*)	X ^(*)	X	X	40000689	13.5
P18	Cover plate	Rp 1 1/4	-	X	X	X	X	18075627	13
P21	Tread-proof, split, with profile seals and angle iron mounting frame (type A 560) for 500 x 500 mm pits (For dual-pump stations with Y-pipe 2 cover plates are fitted next to each other.)								
	A 25 B drain hose set	Rp 1 1/4	-	X ^(*)	-	X ^(*)	-	18079719	3
	Comprising: rigid coupling with external thread, 6 m plastic hose DN 25, rapid-action hose coupling Rp 1 1/4, (free passage 21 mm)	C 42	-	-	-	-	-	42209411	1.7
	Additional part – see P32								

9 Expander or reducer required

Item	Description	Connection	Immersion depth [m]	AmaDrainer				Mat. No.	[kg]
				4../10	4../35	5../10K	522/11		
 P22 Hose connection set Consisting of: 1 hose nozzle with external thread (PVC), 1 hose clamp (CrNiMo steel) For additional parts see P32 and P33		C 52-G 11/2	-	X	X	-	-	19072025	0.2
		C 52-G 2	-	-	-	X	X ^(*)	18040259	0.2
		B 75-G 2	-	-	-	X	X ^(*)	18040205	0.2
 P24 Storz rigid coupling with internal thread to DIN ISO 228/1 Aluminium alloy, required pipe part see P32		C-G 11/2	-	X	X	-	-	01002463	0.3
		C-G 2	-	-	-	X	X	00520120	0.3
		B-G 11/2	-	X	X	-	-	01062591	0.1
		B-G 2	-	-	-	X	X	00133084	0.4
 P24 Storz rigid coupling with external thread to DIN ISO 228/1 Aluminium alloy		C 52 / G 2	-	-	-	-	X	00524370	0.2
 P26 Storz hose coupling		C 52 (DIN 14321)	-	X	X	-	-	00524551	0.3
		B 75 (DIN 14322)	-	-	-	X	X	00520454	0.7
 P27 Hose clip DIN 3017, chrome steel		AL 40-60 C (DIN 3017)	-	X	X	-	-	00114522	0.02
		AL 70-90 B (DIN 3017)	-	-	-	X	X	01063363	0.03
 P28 Synthetic hose DN 40, with integrated C couplings, DIN 14811 Synthetic hose DN 50, DIN 14811, with integrated C couplings Synthetic hose DN 75, with integrated B couplings, DIN 14811 Synthetic hose DN 75, supplied by the metre, without couplings (max. 30 m), DIN 14811		C 42-5 m	-	X	X	X	X	01062592	1.7
		C 42-10 m	-	X	X	X	X	01062593	2.8
		C 42-20 m	-	X	X	X	X	01062594	5
		C 52-5 m	-	X	X	X	X	00522262	2.3
		C 52-10 m	-	X	X	X	X	00522263	4.2
		C 52-20 m	-	X	X	X	X	00522264	5.7
		B 75, 20 m	-	-	X	X	X	00522265	10
		B 75, per metre	-	-	X	X	X	00540104	0.3
 P29 Threaded flange for Y-pipe (P13), internal thread Threaded flange for guide hoop arrangement (P2) and guide wire arrangement (P4) Comprising flange, 4 hexagon head bolts with nuts and washers and 1 sealing element		DN 40 / Rp 11/2	-	X	X	X ^(*)	X ^(*)	00260478	1.8
		DN 50/Rp 2	-	X ^(*)	X ^(*)	X	X	00260479	2.5
		DN 50 / Rp 2	-	-	-	-	X	19551353	3.23
 P32 Hexagon nipple for Storz rigid coupling C (P24), EN 10242 Galvanised steel, external/external thread Hexagon nipple for Storz rigid coupling C and B (P24) and for hose connection set (P22) Galvanised steel, external/external thread		R 11/2	-	X	X	-	-	00240874	0.5
		R 2	-	-	-	-	X	00240876	0.6
 P32 Pipe extension for Storz rigid coupling B (P24) 1.4401, external/external thread Pipe extension for Storz rigid coupling C and B (P24) Galvanised steel, external/external thread		ET 11/2 / ET R 11/2 x 310	-	X	X	-	-	11037771	1
		ET 2 / ET R 2 x 150	-	-	-	X	-	00250494	0.6
 P33 Reducer DN 32 EN 10242, galvanised steel, external/internal thread for A 25 B drain hose set (P21) Reducer DN 40 EN 10242, galvanised steel, external/internal thread for A 25 B drain hose set (P21)		ET R 11/2 / IT Rp 11/4	-	X ^(*)	-	-	-	00240679	0.2
		ET R 2 / IT Rp 11/4	-	-	-	X ^(*)	X ^(*)	00240680	0.4
		ET R 2 / IT Rp 11/2	-	-	-	X ^(*)	X ^(*)	00240686	0.5
 P33 Expander - nominal size, socket DN 50, EN 10242, galvanised steel For additional parts see P32 and P33 Nominal diameter increase, socket DN 65, EN 10242, galvanised steel For additional parts see P32 and P33		ET R 11/2 / IT Rp 2	-	X ^(*)	X ^(*)	-	-	00241778	0.38
		IT Rp 2 / IT Rp 2 1/2	-	-	-	X ^(*)	X ^(*)	40982361	1.3
 P35 Cooling jacket retrofit kit Comprising cooling jacket, connection socket with internal thread Rp2 and two O-rings, to retrofit AmaDrainer 4../10 to the same design, AmaDrainer 5../10 K The piping of pumps already installed must be adjusted accordingly.		-	-	X	-	-	-	18040775	0.5

Control units and switchgear

Not valid for France.

Table 18: Overview of control units and switchgear

Item	Description	230 V	400 V	Type	Strom Min. / Max. [A]	AmaDrainer										Mat. No.	[kg]
						1~				3~							
						.05.E	.07.E	.11.E	.15.E	.05.D	.07.D	.11.D	.15.D	.22.D	522/11		
	MSE switchgear	X	-	40.1	2,5 - 4,0	X	-	-	-	-	-	-	-	-	19070137	1	
	for Float switch	X	-	60.1	4,0 - 6,0	-	X	-	-	-	-	-	-	-	19070138	1	
	With integrated motor protection relay, thermal motor protection, manual-O-automatic selector switch, operation and fault indicator lamps Dimensions (W x H x D) 100 x 170 x 112 mm	X	-	80.1	5,5 - 8,0	-	-	X	-	-	-	-	-	-	19070139	1	
		X	-	100.1	8,0 - 11,5	-	-	-	X	-	-	-	-	-	19070140	1	
	MSD switchgear	-	X	16.1	1,2 - 1,8	-	-	-	-	X	-	-	-	-	19070114	1	
	Float switch	-	X	25.1	1,8 - 2,6	-	-	-	-	-	X	X	-	-	19070115	1.58	
	With integrated motor protection relay, thermal motor protection, manual-O-automatic selector switch, operation and fault indicator lamps Dimensions (W x H x D) 100 x 170 x 112 mm	-	X	40.1	2,5 - 4,0	-	-	-	-	-	-	X	-	-	19070116	1	
		-	X	60.1	3,7 - 5,5	-	-	-	-	-	-	-	X	X	19070117	1.55	
E 3	CEE motor protection plug, type Hyper ⁽¹⁰⁾ 3/N/PE 16 A, IP X4 Phase inverter, contactor up to 4 kW, manual-O-automatic selector switch, connections for three-phase motor, thermal circuit breaker and float switch	-	X	Hyper CEE	-	-	-	-	-	X	X	X	X	X	X	18040512	0.8
	Multi-functional plug, type Hyper, with motor protection relay	-	X	Hyper 18.1	1,2 - 1,8	-	-	-	-	X	-	-	-	-	-	19071490	1
	3/N/PE 16 A, IP X4, phase inverter, motor monitoring, contactor up to 4 kW, motor protection relay, manual-O-automatic selector switch, reset key, indicator lamps for direction of rotation, operation and fault, connections for three-phase motor, thermal circuit breaker and float switch	-	X	Hyper 26.1	1,8 - 2,6	-	-	-	-	-	X	X	-	-	-	19071491	1
		-	X	Hyper 37.1	2,6 - 3,7	-	-	-	-	-	-	-	X	-	-	19071492	0.98
		-	X	Hyper 55.1	3,7 - 5,5	-	-	-	-	-	-	-	-	X	X	19071493	0.98
	LevelControl Basic 2 control unit for single-pump station, IP54	X	-	BC1 230 ^{DFNO} 100	Up to 10	X	X	X	X	-	-	-	-	-	-	19073760	4.5
	DOL starting, with manual-O-automatic selector switch, indicator lamps and control panel, high water alert, integrated alarm buzzer 85 dB(A), operating hours counter/start-stop cycles per pump, voltage measurement, phase monitoring, indication of water level, volt-free contact for general fault message, motor temperature warning (thermal circuit breaker) – self-acknowledging; motor leakage/moisture monitoring, optional mains-independent alarm via rechargeable battery (E90) For float switch or 4 - 20 mA sensor, optionally with master switch, 400 x 281 x 135 mm 400 V variant with motor protection switch 230 V variant with external socket	-	X	BC1 400 ^{DFNO} 025	1,6 - 2,5	-	-	-	-	X	X	-	-	-	-	19073762	4.5
		-	X	BC1 400 ^{DFNO} 040	2,5 - 4,0	-	-	-	-	-	-	X	X	-	-	19073763	4.67
	-	X	BC1 400 ^{DFNO} 063	4,0 - 6,3	-	-	-	-	-	-	-	-	X	X	19073764	4.5	

10 Required for AmaDrainer ND only

Item	Description	230 V	400 V	Type	Strom Min. / Max. [A]	AmaDrainer										Mat. No.	[kg]
						1~				3~							
						.05.E	.07.E	.11.E	.15.E	.05.D	.07.D	.11.D	.15.D	.22.D	522/11		
	LevelControl Basic 2 control unit for dual-pump station, IP54	X	-	BC2 230 ^{DFNO} 100	Up to 10	X	X	X	X	-	-	-	-	-	-	19073774	4.7
		-	X	BC2 400 ^{DFNO} 025	1,6 - 2,5	-	-	-	-	X	X	-	-	-	-	19073776	4.7
	Peak load operation, stand-by pump, DOL starting, with manual-0-automatic selector switch, indicator lamps and control panel, high water alert, integrated alarm buzzer 85 dB(A), operating hours counter/start-stop cycles per pump	-	X	BC2 400 ^{DFNO} 040	2,5 - 4,0	-	-	-	-	-	-	X	X	-	-	19073777	4.7
	Voltage measurement, phase monitoring, volt-free contact for general fault message Optional mains-independent alarm via re-chargeable battery with charge controller (E90) For float switch or 4 - 20 mA sensor, optionally with master switch, 400 x 281 x 135 mm 400 V variant: with motor protection switch 230 V variant: with external socket	-	X	BC2 400 ^{DFNO} 063	4,0 - 6,3	-	-	-	-	-	-	-	-	X	X	19073778	4.7
Optional components for LevelControl ^{III}																	
O1 	Master switch for LevelControl Basic 2 BC, fitted 3-pole, 20 A, lockable	-	-	-	-	X	X	X	X	X	X	X	X	X	X	01143084	0.19

 Optional components are not EDI-compatible (configurable range)

Operation with mini control system

Order separate float switches and appropriate cable lengths with Ama-Drainer 405 NE to 415 NE and 505 NE to 515 NE.

LevelControl with float switch(es)

Single pump:

- At least 1 float switch for pump On/Off
- At least 2 float switches for pump On/Off and high water alert

Dual pump:

- At least 2 floats for pump On/Off
- At least 3 float switches for pump On/Off and high water alert

Twin operation with two level switches at different levels

If two pumps are used in the same location, we recommend operating them via the LevelControl unit. This enables automatic alternating, peak load and stand-by operation. External alarm switchgear will not be required as LevelControl features an integrated alarm function.

Connection to the control station

With the exception of MSE, MSD and Hyper, each control unit features a volt-free contact for transmitting the general fault message to the control station.

Control units and switchgear

Only valid for France.

Table 19: Overview of control units and switchgear for France

Item	Description	230 V	400 V	Type	Strom Min. / Max.	AmaDrainer										Mat. No.	[kg]
						1~				3~							
					[A]	.05.E	.07.E	.11.E	.15.E	.05.D	.07.D	.11.D	.15.D	.22.D	522/11		
	LevelControl Basic 2 control unit for single-pump station, IP 54	✗	-	BC1 230 ^{DFNM} 040 02	2,5 - 4,0	✗	-	-	-	-	-	-	-	-	-	19073873	4.5
	In compliance with NF C 15-100	✗	-	BC1 230 ^{DFNM} 063 02	4,0 - 6,3	-	✗	-	-	-	-	-	-	-	-	19073874	4.5
	DOL starting With manual-0-automatic selector switch, indicator lamps and control panel	✗	-	BC1 230 ^{DFNM} 100 02	6,3 - 10,0	-	-	✗	✗	-	-	-	-	-	-	19073875	4.5
	High water alert Integrated alarm buzzer, 85 dB(A)	-	✗	BC1 400 ^{DFNO} 025 02	1,6 - 2,5	-	-	-	-	✗	✗	-	-	-	-	19073877	4.5
	Operating hours counter / counter for start-stop cycles per pump	-	✗	BC1 400 ^{DFNO} 040 02	2,5 - 4,0	-	-	-	-	-	-	✗	✗	-	-	19073878	5.01
	Voltage measurement, phase monitoring Volt-free contact for general fault message Optional mains-independent alarm via re- chargeable battery	-	✗	BC1 400 ^{DFNO} 063 02	4,0 - 6,3	-	-	-	-	-	-	-	-	✗	✗	19073879	4.5
	LevelControl Basic 2 control unit for dual-pump station, IP54	✗	-	BC2 230 ^{DFNM} 040 02	2,5 - 4,0	✗	-	-	-	-	-	-	-	-	-	19073883	4.7
	In compliance with NF C 15-100	✗	-	BC2 230 ^{DFNM} 063 02	4,0 - 6,3	-	✗	-	-	-	-	-	-	-	-	19073884	4.7
	Peak load operation function DOL starting With manual-0-automatic selector switch indicator lamps and control panel	✗	-	BC2 230 ^{DFNM} 100 02	6,3 - 10,0	-	-	✗	✗	-	-	-	-	-	-	19073885	4.7
	High water alert Integrated alarm buzzer, 85 dB(A)	-	✗	BC2 400 ^{DFNO} 025 02	1,6 - 2,5	-	-	-	-	✗	✗	-	-	-	-	19073887	4.7
	Operating hours counter / counter for start-stop cycles per pump	-	✗	BC2 400 ^{DFNO} 040 02	2,5 - 4,0	-	-	-	-	-	-	✗	✗	-	-	19073888	4.7
	Voltage measurement, phase monitoring Volt-free contact for general fault message Optional mains-independent alarm via re- chargeable battery	-	✗	BC2 400 ^{DFNO} 063 02	4,0 - 6,3	-	-	-	-	-	-	-	-	✗	✗	19073889	5.63
	CEE motor protection plug, type Hyper ⁽²⁾ 3/N/PE 16 A, IP X4 Phase inverter, contactor up to 4 kW, manual-0- automatic selector switch, connections for three-phase motor, thermal circuit breaker and float switch	-	✗	Hyper CEE	-	-	-	-	-	✗	✗	✗	✗	✗	✗	18040512	0.8
	Multi-functional plug, type Hyper, with motor protection relay	-	✗	Hyper 18.1	1,2 - 1,8	-	-	-	-	✗	-	-	-	-	-	19071490	1
		-	✗	Hyper 26.1	1,8 - 2,6	-	-	-	-	-	✗	✗	-	-	-	19071491	1
	3/N/PE 16 A, IP X4, phase inverter, motor moni- toring, contactor up to 4 kW, motor protection re- lay, manual-0-automatic selector switch, reset key, indicator lamps for direction of rotation, op- eration and fault, connections for three-phase motor, thermal circuit breaker and float switch	-	✗	Hyper 37.1	2,6 - 3,7	-	-	-	-	-	-	-	✗	-	-	19071492	0.98
		-	✗	Hyper 55.1	3,7 - 5,5	-	-	-	-	-	-	-	-	✗	✗	19071493	0.98

Optional components are not EDI-compatible (configurable range)

Operation with mini control system

Order separate float switches and appropriate cable lengths with Ama-Drainer 405 NE to 415 NE and 505 NE to 515 NE.

LevelControl with float switch(es)

Single pump:

- At least 1 float switch for pump On/Off
- At least 2 float switches for pump On/Off and high water alert

Dual pump:

- At least 2 floats for pump On/Off
- At least 3 float switches for pump On/Off and high water alert

Twin operation with two level switches at different levels

If two pumps are used in the same location, we recommend operating them via the LevelControl unit. This enables automatic alternating, peak load and stand-by operation. External alarm switchgear will not be required as LevelControl features an integrated alarm function.

Connection to the control station

With the exception of MSE, MSD and Hyper, each control unit features a volt-free contact for transmitting the general fault message to the control station.

LevelControl Basic 2 control units

 Not valid for France.

Table 20: Symbols key

Symbol	Description
o	Optional
x	Available
-	Not available

Table 21: Overview of LevelControl Basic 2 control units

Feature	Single-pump station Float switches incl. 4-20 mA	Dual-pump station Float switches incl. 4-20 mA
230 V, up to 10.0 A	BC1 230 _{DFNO} 100	BC2 230 _{DFNO} 100
400 V, 1,6 - 2,5 A	BC1 400 _{DFNO} 025	BC2 400 _{DFNO} 025
400 V, 2,5 - 4,0 A	BC1 400 _{DFNO} 040	BC2 400 _{DFNO} 040
400 V, 4,0 - 6,3 A	BC1 400 _{DFNO} 063	BC2 400 _{DFNO} 063
Functions		
Draining	x	x
Filling using float switch(es)	x	x
Stand-by pump: 1 pump redundant	-	x
Pump changeover after each start	-	x
Pump changeover in event of pump fault	-	x
Peak load operation function	-	x
Runtime limitation	x	x
OFF via after-run time	x	x
OFF via level	x	x
Functional check run after idle period	x	x
Alert history	-	-
Display and operation		
7-segment display	x	x
Indication of water level	Switching points	Switching points
Operation / fault / pump running (displayed for each pump)	Multicolour LED	Multicolour LED
General fault (traffic light)	LED	LED
High water	LED	LED
Mains voltage	x	x
Mains frequency	-	-
Motor current per pump	-	-
Operating hours of each pump	x	x
Operating hours of the system	-	-
Starts per pump	x	x
Effective power per pump	-	-
Rotary field recognition of mains power supply	x	x
Phase monitoring	x	x
Change of switching levels via control panel	-	-
Housing H x W x D [mm], IP54		
Plastic 400 x 281 x 135	x	x
Sheet steel 400 x 300 x 155	-	-
Sheet steel 600 x 400 x 200	-	-
Built-in components		
Master switch (lockable)	o	o
Manual-O-automatic selector switch per pump	x	x
DOL starting	x	x
Star-delta starting	-	-
Shockproof socket 230 V	For 230 V	For 230 V
Motor protection		
Overcurrent protective device per pump	For 230 V	For 230 V
Motor protection switch per pump (overcurrent protection and short-circuit protection)	At 400 V	At 400 V

Feature	Single-pump station Float switches incl. 4-20 mA	Dual-pump station Float switches incl. 4-20 mA
Motor temperature warning input	X	X
Motor temperature alert input	X	X
Pump		
Thermal circuit breaker (TCB) / bimetal switch per pump	400 V: brought out	400 V: brought out
Optional components		
Rechargeable battery for powering the device	O	O
Alarm equipment		
1 free alarm input	X	X
1 digital input for high water alert	X	X
Volt-free contact (changeover contact) for general fault message / "in operation" message	X	X
Piezo buzzer 85 dB(A)	X	X
Horn / alarm combination / alarm strobe light 12 V DC	O	O
Inputs/outputs		
Inputs for float switches	4	4
4-20 mA analog input	X	X
Integrated pneumatic pressure sensor up to 3 metres of water (up to 10 metres on request)	-	-
Bubbler control with compressor up to 2 metres of water	-	-
Remote acknowledgement	X	X
12 V DC connection for horn, etc.	X	X
Sensors		
Float switch (NO contact)	O	O
F1 leakage sensor	O	O
Tools		
KSB ServiceTool	O	O

LevelControl Basic 2 control units

 Only valid for France.

Table 22: Symbols key

Symbol	Description
o	Optional
x	Available
-	Not available

Table 23: Overview of LevelControl Basic 2 control units

Feature	Single-pump station Float switches incl. 4-20 mA	Dual-pump station Float switches incl. 4-20 mA
230 V, 2,5 - 4,0 A	BC1 230 ^{DFNM} 040 02	BC2 230 ^{DFNM} 040 02
230 V, 4,0 - 6,3 A	BC1 230 ^{DFNM} 063 02	BC2 230 ^{DFNM} 063 02
230 V, 6,3 - 10,0 A	BC1 230 ^{DFNM} 100 02	BC2 230 ^{DFNM} 100 02
400 V, 1,6 - 2,5 A	BC1 400 ^{DFNO} 025 02	BC2 400 ^{DFNO} 025 02
400 V, 2,5 - 4,0 A	BC1 400 ^{DFNO} 040 02	BC2 400 ^{DFNO} 040 02
400 V, 4,0 - 6,3 A	BC1 400 ^{DFNO} 063 02	BC2 400 ^{DFNO} 063 02
Functions		
Draining	x	x
Filling using float switch(es)	x	x
Stand-by pump: 1 pump redundant	-	x
Pump changeover after each start	-	x
Pump changeover in event of pump fault	-	x
Peak load operation function	-	x
Runtime limitation	x	x
OFF via after-run time	x	x
OFF via level	x	x
Functional check run after idle period	x	x
Alert history	-	-
Display and operation		
7-segment display	x	x
Indication of water level	Switching points	Switching points
Operation / fault / pump running (displayed for each pump)	Multicolour LED	Multicolour LED
General fault (traffic light)	LED	LED
High water	LED	LED
Mains voltage	x	x
Mains frequency	-	-
Motor current per pump	-	-
Operating hours of each pump	x	x
Operating hours of the system	-	-
Starts per pump	x	x
Effective power per pump	-	-
Rotary field recognition of mains power supply	x	x
Phase monitoring	x	x
Change of switching levels via control panel	-	-
Housing H × W × D [mm], IP54		
Plastic 400 × 281 × 135	x	x
Sheet steel 400 × 300 × 155	-	-
Sheet steel 600 × 400 × 200	-	-
Built-in components		
Master switch (lockable)	x	x
Manual-O-automatic selector switch per pump	x	x
DOL starting	x	x
Star-delta starting	-	-
Shockproof socket 230 V	For 230 V	For 230 V
Motor protection		

Feature	Single-pump station Float switches incl. 4-20 mA	Dual-pump station Float switches incl. 4-20 mA
Motor protection switch per pump (overcurrent protection and short-circuit protection)	X	X
Motor temperature warning input	X	X
Motor temperature alert input	X	X
Pump		
Thermal circuit breaker (TCB) / bimetal switch per pump	400 V: brought out	400 V: brought out
Optional components		
Rechargeable battery for powering the device	O	O
Alarm equipment		
1 free alarm input	X	X
1 digital high water alert input (e.g. for float switch)	X	X
Volt-free contact (changeover contact) for general fault message / "in operation" message	X	X
Piezo buzzer 85 dB(A)	X	X
Horn / alarm combination / alarm strobe light 12 V DC	O	O
Inputs/outputs		
Inputs for float switches	4	4
4-20 mA analog input	X	X
Integrated pneumatic pressure sensor up to 3 metres of water (up to 10 metres on request)	-	-
Bubbler control with compressor up to 2 metres of water	-	-
Remote acknowledgement	X	X
12 V DC connection for horn, etc.	X	X
Sensors		
Float switch (NO contact)	O	O
F1 leakage sensor	O	O
Tools		
KSB ServiceTool	O	O

Alarm switchgears for pumps, non-ATEX-compliant

Table 24: AS0 / AS1 / AS2 / AS4 / AS5

Item	Description	Mat. No.	[kg]
E50 	Alarm switchgear AS0 With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch, F1 leakage sensor (item E64), M1 alarm contactor or signal relay of control unit as contactor.	29128401	0.5
E51 	Alarm switchgear AS2 With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp, volt-free contact for hook-up to a control station Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch, F1 leakage sensor (item E64) or signal relay of control unit as contactor.	29128422	0.5
E52 	Alarm switchgear AS4 With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp, volt-free contact for hook-up to a control station, self-charging power supply unit for 5 hours of operation in the event of a power failure Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch (E60), F1 leakage sensor (item E64) or signal relay of control unit as contactor.	29128442	0.5
E53 	Alarm switchgear AS5 Mains-independent, with self-charging power supply unit for 10 hours of operation in the event of a power failure, mains pilot LED, fault indicator light, acknowledgement button, volt-free contact for hook-up to a control station, ready for connection with 1.8 m power cable and plug. ISO housing, IP41, H x W x D = 165 x 190 x 75 [mm]. Use float switch (E60) or signal relay of control unit as contactor.	00530561	1.7
E55 	Alarm switchgear AS1 In IP30 ISO plug housing, mains-independent, with self-charging power supply unit for 5 hours of operation in the event of a power failure, acoustic signalling device 70 dB(A) with circuit breaker and integrated signal transmitter with 3-metre power cable, max. 60 °C, not suitable for steam and condensate. 1. High water alert by suspending the sensor in a (pump) sump above the pump start-up point. 2. Water alert signal at a water level of only 1 mm by placing the sensor on the floor in areas with a flooding or leakage risk, e.g. the cellar or next to the washing machine in the kitchen or bathroom.	00533740	0.9

Control unit/switchgear accessories
Table 25: Control unit / switchgear accessories

Item	Description	Length of electric cable / flexible tube	AmaDrainer										Mat. No.	[kg]	
			1~				3~								
			[m]	.05.E	.07.E	.11.E	.15.E	.05.D	.07.D	.11.D	.15.D	.22.D			522/11
	Float switch with free cable end Function: circuit closed in upper float position (NO contact) Float switch housing: polypropylene Fluid temperature: ≤ 70 °C Power cable: H07RN-F3G1	3	X	X	X	X	X	X	X	X	X	X	11037742	0.5	
		5	X	X	X	X	X	X	X	X	X	X	11037743	0.74	
		10	X	X	X	X	X	X	X	X	X	X	11037744	1.28	
		15	X	X	X	X	X	X	X	X	X	X	11037745	1.84	
		20	X	X	X	X	X	X	X	X	X	X	11037746	2.25	
		25	X	X	X	X	X	X	X	X	X	X	11037747	2.9	
		30	X	X	X	X	X	X	X	X	X	X	11037748	3.4	
	Float switch with shockproof plug Function: circuit closed in upper float position (NO contact) Float switch housing: polypropylene Fluid temperature: 70 °C max. Power cable: H07RN-F3G1	3	X	X	X	-	-	-	-	-	-	-	11037749	1.1	
		5	X	X	X	-	-	-	-	-	-	-	-	11037750	1.3
		10	X	X	X	-	-	-	-	-	-	-	-	11037751	1.29
		20	X	X	X	-	-	-	-	-	-	-	-	11037752	2.7
	Float switch with free cable end, oil-resistant Function: circuit closed in upper float position (NO contact) Float switch housing: polypropylene Fluid temperature: 70 °C max. Power cable: PUR 3×1.0 mm²	5	X	X	X	X	X	X	X	X	X	X	11037753	0.8	
		10	X	X	X	X	X	X	X	X	X	X	X	11037754	1.2
		20	X	X	X	X	X	X	X	X	X	X	X	11037755	2
	Float switch with free cable end ¹³⁾ Function: circuit open in upper float position (NC contact) Float switch housing: polypropylene Fluid temperature: 70 °C max. Power cable: H07RN-F3G1	5	X	X	X	X	X	X	X	X	X	X	11037756	0.8	
		10	X	X	X	X	X	X	X	X	X	X	X	11037757	1.4
		20	X	X	X	X	X	X	X	X	X	X	X	11037758	2.6
	Float switch with shockproof plug Function: circuit open in upper float position (normally closed contact) Float switch housing: polypropylene Fluid temperature: 70 °C max. Power cable: H07RN-F3G1	3	X	X	X	-	-	-	-	-	-	-	11037759	0.6	
		5	X	X	X	-	-	-	-	-	-	-	-	11037760	0.79
		10	X	X	X	-	-	-	-	-	-	-	-	11037761	1.5
		20	X	X	X	-	-	-	-	-	-	-	-	11037762	2.7
E64	F1 leakage sensor Contact for alarm switchgears AS 0, AS 2, AS 4 or as alarm transmitter for LevelControl Basic 2 Alarm transmission options: High water alert by suspending the sensor in a (pump) sump above the pump start-up point. Warning at a water level of 1 mm in areas with a flooding or leakage risk (e.g. in the basement or next to the washing machine in the kitchen or bathroom) Dimensions [mm]: 52 × 21 × 20 (H × W × D)	3 m	X	X	X	X	X	X	X	X	X	X	19072366	0.2	
	Open pressure bell set, pneumatic and bubbler control with polyamide tube 8 x 1 mm	10	X	X	X	X	X	X	X	X	X	X	19071721	1.2	
		20	X	X	X	X	X	X	X	X	X	X	19071837	2	
E66	Closed pressure bell set, pneumatic with polyamide tube 8 x 3 mm	10	X	X	X	X	X	X	X	X	X	X	19071722	3.5	

Item	Description	Length of electric cable / flexible tube	AmaDrainer										Mat. No.	[kg]
			1~				3~							
		[m]	.05.E	.07.E	.11.E	.15.E	.05.D	.07.D	.11.D	.15.D	.22.D	522/11		
E70 	Horn, 12 V DC, 105 dB, 150 mA, IP54 Suitable for indoor installation and outdoor installation. Protect against moisture.	-	X	X	X	X	X	X	X	X	X	01086547	0.09	
E80 	Safety switch STECKMAT trips in approx. 0.03 seconds, from approx. 0.03 A 230 V / 10 A	-	X	X	X	X	-	-	-	-	-	00534217	0.5	
E90 	Rechargeable battery retrofit kit for LevelControl Basic 2, type BC Scope of supply: 2 rechargeable batteries (6 V, 1.3 Ah) and charge controller	-	X	X	X	X	X	X	X	X	X	19074194	0.8	

Cross sectional drawings with list of components

General assembly drawing of AmaDrainer 4../10, 4../35, 5../10K

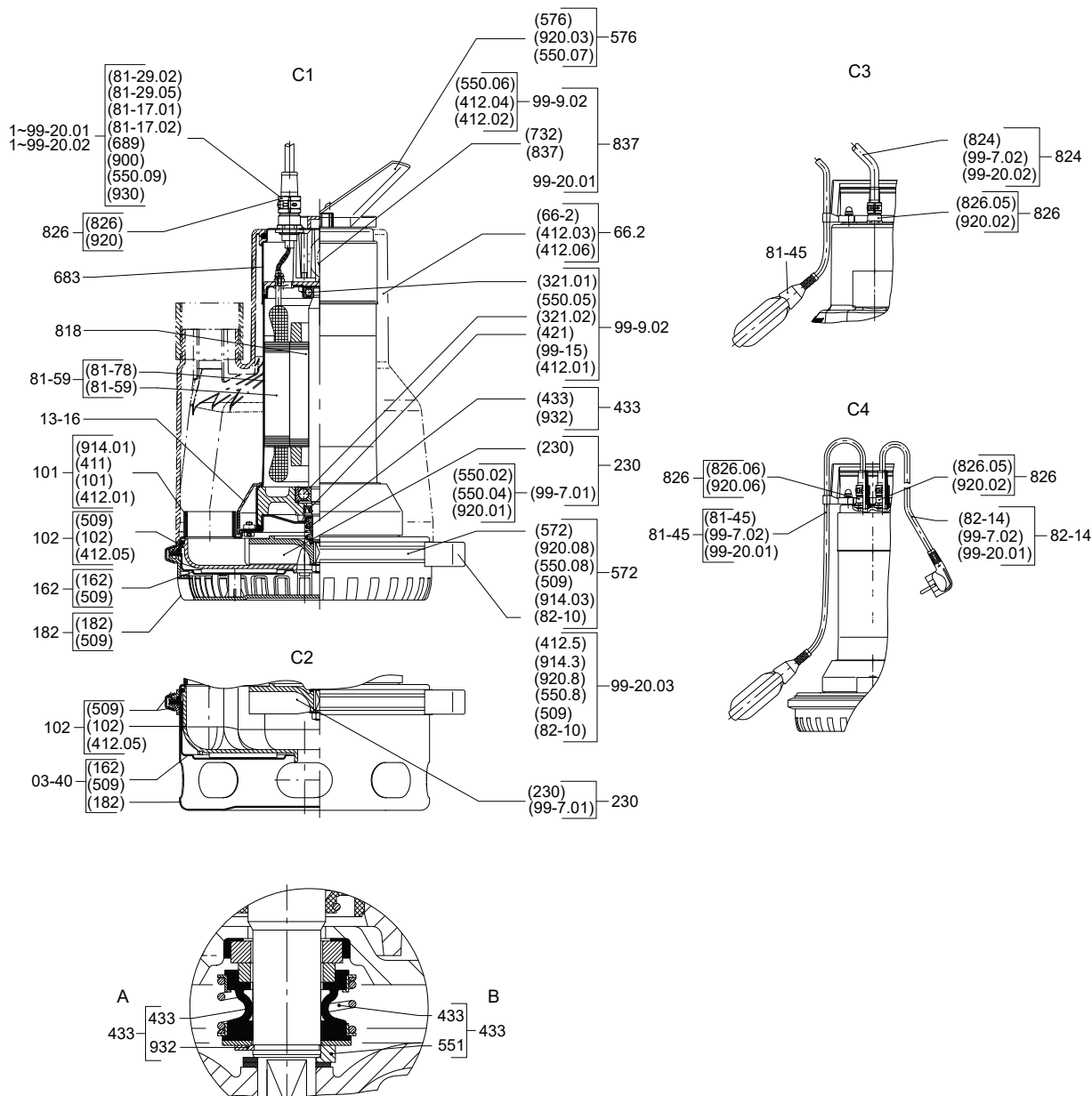


Fig. 15: General assembly drawing of AmaDrainer 4../10, 4../35, 5../10K

A	Material variant A / R	B	Material variant C
C1	AmaDrainer 4../10, 4../35, 5../10K	C2	AmaDrainer 4../35
C3	AmaDrainer SD/..	C4	AmaDrainer SE

Table 26: List of components

Part No.	Description	Comprising
03-40	Pump foot / suction cover assembly (for .../35)	Suction cover 162 Pump foot 182 Intermediate ring 509
101	Pump casing, complete	Pump casing 101 Joint ring 411 O-ring 412.01 Hexagon socket head cap screw 914.01
102	Volute casing	Volute casing 102 O-ring 412.05 Intermediate ring 509
13-16	Protective casing	Protective casing 13-16
162	Suction cover	Suction cover 162 Intermediate ring 509
182	Pump foot	Pump foot 182 Intermediate ring 509
230	Impeller, complete	Impeller 230 Impeller installation kit 99-7.01
433	Mechanical seal, complete	Mechanical seal 433 Spacer disc 551 (for variant C only) Circlip 932
572	Clamp, complete	Clamp 572 Intermediate ring 509 Disc 550.08 Cover strip 82-10 Hexagon socket head cap screw 914.03 Nut 920.08
576	Handle, complete	Handle 576 Disc 550.07 Nut 920.03 Label/plate 970
66-2	Cooling jacket, set of accessories	Cooling jacket 66-2 O-ring 412.03/.06
683	Hood	Hood 683
81-45	Float switch (for single-phase unit)	Float switch (6 A / 0.5 m) 81-45 Float switch (10 A / 0.5 m) 81-45 Cable repair kit (single-phase) 99-20.01 Installation kit for hood 99-7.02
81-45	Float switch (for three-phase unit)	Float switch (6 A / 10 m) 81-45
81-59	Stator, complete	Stator 81-59 Stator case 81-78
818	Pump rotor	Pump rotor 818
82-14	Cable with plug (single-phase units)	Cable with plug (3×1 mm ² , length 10 m) 82-14 Cable repair kit (single-phase) 99-20.01 Installation kit for hood 99-7.02
824	Cable (three-phase units)	Cable (6×1 mm ² , length 10 m) 824 Repair kit 99-20.02 Installation kit for hood 99-7.02
826	Cable gland	Cable gland 826 Nut (M20x1.5) 920.05
837	Capacitor (for single-phase unit only)	Capacitor 837 Capacitor holder 732 Cable repair kit (single-phase) 99-20.01 Installation kit for hood 99-7.02
99-7.01	Impeller installation kit	Adjusting washer 550.02 Disc 550.04 Nut 920.01
99-7.02	Installation kit for hood	O-ring 412.02/.04

Part No.	Description	Comprising
99-7.02	Installation kit for hood	Disc 550.06
99-11	Bearing	Deep groove ball bearing 321.01/.02
		O-ring 412.01
		Shaft seal ring 421
		Disc 550.05
		Lubricating oil 99-15
99-20.01/.02	Cable repair kit	Insulation tube 689
		Disc 550.09
		End connector 81-17.01/.02
		Terminal 81-29.02
		Screw 900
		Serrated lock washer 930
99-20.03	Hydraulic system repair kit	O-ring 412.05
		Intermediate ring 509
		Disc 550.08
		Cover strip 82-10
		Hexagon socket head cap screw 914.03
		Nut 920.08

General assembly drawing of AmaDrainer 522 SD/11

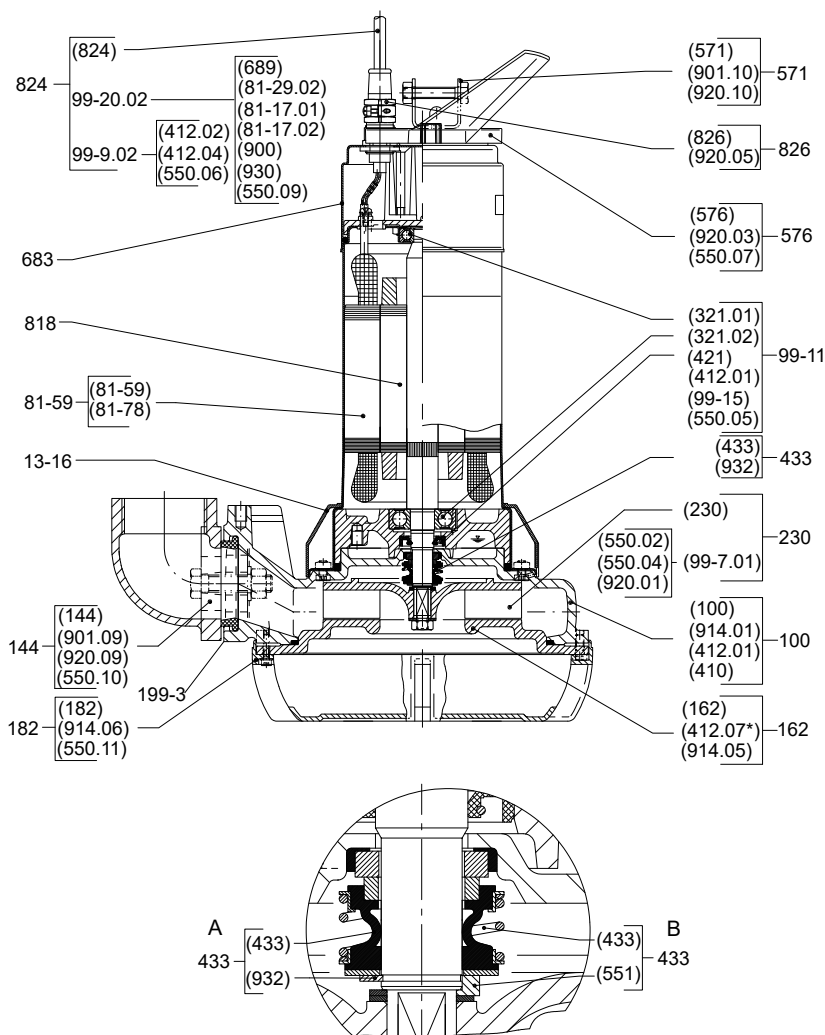


Fig. 16: General assembly drawing of AmaDrainer 522 SD/11

A	Material variant A / R	B	Material variant C
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Table 27: List of components

Part No.	Description	Comprising
100	Casing, complete	Casing 100
		Profile seal 410
		O-ring 412.01
		Hexagon socket head cap screw 914.01
144	Discharge elbow, complete	Discharge elbow 144
		Disc 550.10
		Hexagon head bolt 901.09
		Nut 920.09
13-16	Protective casing	Protective casing 13-16
162	Suction cover	Suction cover 162
		O-ring 412.07
		Hexagon socket head cap screw 914.05
182	Pump foot	Pump foot 182
		Disc 550.11
		Hexagon socket head cap screw 914.06
199-3	Flange adapter ¹⁴⁾	Flange adapter (DN 50) 182.5
		Profile seal 410.02
		Disc 550.12
		Stud 902.01
230	Impeller, complete	Impeller 230
		Impeller installation kit 99-7.01
433	Mechanical seal, complete	Mechanical seal 433
		Spacer disc 551 (for variant C only)
		Circlip 932
571	Bracket, complete	Bracket 571
		Hexagon head bolt 901.10
		Hexagon nut 920.10
576	Handle, complete	Handle 576
		Disc 550.07
		Nut 920.03
683	Hood	Hood 683
81-45	Float switch (for three-phase unit)	Float switch (6 A / 10 m) 81-45
81-59	Stator, complete	Stator 81-59
		Stator case 81-78
818	Pump rotor	Pump rotor 818
824	Cable (three-phase units)	Cable (6×1 mm ² , length 10 m) 824
		Installation kit for hood 99-7.02
		Repair kit 99-20.02
826	Cable gland	Cable gland 826
		Nut (M20×1.5) 920.05
99-7.01	Impeller installation kit	Adjusting washer 550.02
		Disc 550.04
		Nut 920.01
99-7.02	Installation kit for hood	O-ring 412.02/.04
		Disc 550.06
99-11	Bearing	Deep groove ball bearing 321.01/.02
		O-ring 412.01
		Shaft seal ring 421
		Disc 550.05
		Lubricating oil 99-15
99-20.02	Cable repair kit	Insulation tube 689
		Disc 550.09
		End connector 81-17.01/.02
		Terminal 81-29.02
		Screw 900

14 Not shown in drawing

Part No.	Description	Comprising
99-20.02	Cable repair kit	Serrated lock washer 930
99-20.03	Hydraulic system repair kit	O-ring 412.05
		Intermediate ring 509
		Disc 550.08
		Cover strip 82-10
		Hexagon socket head cap screw 914.03
		Nut 920.08

